

# LOCALIZATION IN THE COMMUTANT, STAGE 10:

the selected Dirac operator.

Tilings, kink currents, contact memory and the He-initialised ReLU network  
as a critical random spectral geometry

9 October 2026

## Abstract

We place the bias-free He-initialised ReLU multilayer perceptron with standard Gaussian input inside the framework of random finite spectral triples (Dirac ensembles), Sahasrabudhe's two-level picture of probabilistic combinatorics at small scales, and Klartag's evolving ellipsoid. The placement has one central object, the *selected Dirac operator*: the weights form a random chiral Dirac chain on the layered neuron set, the input selects a projection of the commutative neuron algebra (its gate pattern), and the network is the resolvent of the Dirac chain compressed by that projection. The partition of input space by the selected projection is the activation tiling; its walls are an isotropic random hyperplane process at every layer (level one), and its overlap kernels are the geometry in which the next layer's walls live (level two, the lift). Proved results: (i) a Crofton formula for the tiling and the folding law at He criticality (lengths preserved, chords contract like  $3\pi/l$ ); (ii) the *kink-current formula*: the exact mean of the network is a sum over walls of the Gaussian surface density of the wall times the downstream transport, equivalently half the expected local time of the pre-activations along a Brownian path; (iii) the third-cumulant memory of every moment chain is a record of contacts: its hub index is the index of a kink and its identity leg the pinned coordinate (Price's theorem), and a frame bound shows that its compressibility is that of its non-Hadamard leg; (iv) the neuron algebra and the modular frame of the layer state are complementary, which is the Hadamard obstruction in operator-algebraic form; (v) the Euler–Stein correction of stage 9 is an average over Gaussian balls of uniformly distributed radius in the tiling, its constant is  $1/(d+2)$  for a defect of Hermite contact order  $d$ , which is the order of the concentration-function coefficient at the kink that the estimator gets wrong; (vi) neuron-pair correlations are Mehler factors, i.e. imaginary modular times of the Gibbs state of the Hermite number operator, and the depth flow of the pair temperature is marginal at linear order with a  $\beta^{3/2}$  term produced by the kink, which fixes the exponent of the folding law; (vii) the mean is the unique solution of a positivity-plus-Stein moment problem whose first loop equation is the kink-current formula, and the cumulant chains are its truncations without positivity. (viii) in the simplest framing (input slots as vertices, layers as quadratic forms, tokens as Hermite quanta) the mean of every neuron is twice the trace of its quadratic form, layer two's form is the graph of two-step walks through the hidden layer, cumulants are walk sums on the hidden codegree graph, the Gaussian closure is the two-token walk with exact contacts, and the third-cumulant memory is first-order exchange whose hub coefficients are themselves walks; (ix) the quantum-cut (Kikuchi) sparsifiers of Basu–Brakensiek–Kothari–Puterman are Schur–Weyl duality in action, the network carries the bosonic counterpart (Howe duality, with the radial foliation) and the hard-core one (gates), the memory has no redundancy (every hub a bridge), and a higher tier with random dynamics acts on the gates through wall currents: the gate covariance is the integrated Gram of wall currents, localization freezes deep gates first, and the two tiers form a non-commuting square. Every statement is labelled as proved, known (with a reference), or conjectural; the identities are checked by accompanying scripts.

# Contents

|          |                                                                                              |           |
|----------|----------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>The placement in one page</b>                                                             | <b>3</b>  |
| <b>2</b> | <b>Two levels: random walls and their lift</b>                                               | <b>4</b>  |
| 2.1      | Level one: the walls of a layer are an isotropic point process . . . . .                     | 4         |
| 2.2      | Crofton: how many walls a curve crosses . . . . .                                            | 5         |
| 2.3      | The lift at He criticality: lengths are preserved, chords contract . . . . .                 | 5         |
| 2.4      | The lift as a tower of algebras . . . . .                                                    | 6         |
| <b>3</b> | <b>The network as a random spectral triple</b>                                               | <b>6</b>  |
| 3.1      | The chiral Dirac chain . . . . .                                                             | 6         |
| 3.2      | The network is a compressed resolvent . . . . .                                              | 7         |
| 3.3      | The selected Dirac operator and the induced ensemble . . . . .                               | 7         |
| 3.4      | Distance on states . . . . .                                                                 | 8         |
| <b>4</b> | <b>The mean is a kink current</b>                                                            | <b>9</b>  |
| 4.1      | The kink-current formula . . . . .                                                           | 9         |
| 4.2      | What an estimator has to compute: conditional expectations on walls . . . . .                | 10        |
| <b>5</b> | <b>Estimators are unsharp selections; their memory is a record of contacts</b>               | <b>10</b> |
| 5.1      | Moment chains are unsharp selections . . . . .                                               | 10        |
| 5.2      | Pinning and births: the hub is a kink . . . . .                                              | 11        |
| 5.3      | Kinks as points: codegree and the frame bound . . . . .                                      | 12        |
| 5.4      | Complementarity of the neuron algebra and the layer state . . . . .                          | 12        |
| 5.5      | Parity: the odd sector and broken symmetry . . . . .                                         | 13        |
| <b>6</b> | <b>The clock of the selection: Mehler factors, modular time and the Euler–Stein constant</b> | <b>14</b> |
| 6.1      | The Euler–Stein correction probes the estimator on balls of uniform radius . . . . .         | 14        |
| 6.2      | Hermite contact order and the Euler–Stein constant . . . . .                                 | 14        |
| 6.3      | Mehler factors are imaginary modular times . . . . .                                         | 15        |
| <b>7</b> | <b>Bootstrap: the mean as the unique solution of a positivity problem</b>                    | <b>16</b> |
| <b>8</b> | <b>The token picture: vertices, quadratic forms, two-token walks</b>                         | <b>17</b> |
| 8.1      | Dictionary . . . . .                                                                         | 18        |
| 8.2      | Layer one: each neuron is a one-mode token tower, and its mean is a trace . . . . .          | 18        |
| 8.3      | Layer two: the quadratic form is the graph of two-step walks . . . . .                       | 19        |
| 8.4      | How many tokens: depth heats the tower . . . . .                                             | 20        |
| 8.5      | Two tiers: the input law is the environment . . . . .                                        | 20        |
| 8.6      | The two-token walk and its environment . . . . .                                             | 21        |
| 8.7      | What sparsifies, and what does not . . . . .                                                 | 21        |
| <b>9</b> | <b>Kikuchi sparsification, two dualities, and a two-tier dynamics</b>                        | <b>22</b> |
| 9.1      | What the quantum-cut sparsification paper does . . . . .                                     | 22        |
| 9.2      | The network has the same two-duality structure, once bosonic and once hard-core . . . . .    | 23        |
| 9.3      | What transfers: the memory has no redundancy . . . . .                                       | 23        |
| 9.4      | A higher tier with random dynamics . . . . .                                                 | 24        |

|                                                                           |           |
|---------------------------------------------------------------------------|-----------|
| 9.5 The two tiers form a non-commuting square . . . . .                   | 25        |
| <b>10 What the placement explains, what it predicts, and what is open</b> | <b>26</b> |
| 10.1 What the placement explains . . . . .                                | 26        |
| 10.2 What it predicts (falsifiable, theoretical) . . . . .                | 26        |
| 10.3 Open problems . . . . .                                              | 27        |

## 1 The placement in one page

Fix  $W_1, \dots, W_L \in \mathbb{R}^{n \times n}$  with i.i.d.  $N(0, 2/n)$  entries, put  $h_0 = x \sim \gamma = N(0, I_n)$ ,  $z_l = W_l h_{l-1}$ ,  $h_l = \text{relu}(z_l)$ ; the target is  $u = \mathbb{E} h_L$ . The placement consists of four identifications, each of which is a statement proved below.

**1. The space and the metric.** The layered neuron set  $X = \{0, \dots, L\} \times [n]$  carries the commutative algebra  $\mathcal{A} = C(X)$  and the Hilbert space  $\ell^2(X)$ . The weights are one self-adjoint operator  $D = \sum_l (W_l \otimes |l\rangle\langle l-1| + W_l^\top \otimes |l-1\rangle\langle l|)$ , odd for the grading  $(-1)^l$ : a random *chiral Dirac chain*. He initialisation is the Gaussian (quadratic-action) Dirac ensemble  $e^{-\frac{n}{8} \text{Tr} D^2} dD$  at the coupling where the gated transfer has unit mean-square gain (Proposition 3.2).

**2. The selection.** An input  $x$  selects a projection  $P_x \in \mathcal{A}$  (its gate pattern), and the network is the compressed resolvent  $z_L(x) = [(1 - D_\downarrow P_x)^{-1}]_{L0} x$  (Theorem 3.3). The *selected Dirac operator* is  $P_x D P_x$ , the Dirac operator of the active subnetwork. Input space is tiled by the value of the selected Dirac operator (the activation tiling), the Gaussian input induces a probability law on compressions (a Dirac ensemble *induced by the input*), and moving the input across a wall toggles one site, a rank-one contact update of the selected geometry (Proposition 3.5). Localization of the input (stage 9) is the dynamical selection: a Gaussian ball of shrinking radius at a moving centre, whose terminal selected Dirac operator is that of a Gaussian point.

**3. The two levels.** At every layer the walls  $\{z_{l,a} = 0\}$  are the hyperplanes of  $n$  i.i.d. isotropic normals in the previous feature space (level one: a random point process on a sphere). Their overlaps —codegrees of caps, intersections of half-spaces—are arc-cosine kernels on both the input side and the neuron side (Lemma 2.1), and that level-two geometry is the ambient geometry of the next layer’s walls (the lift). Integral geometry (Crofton) counts the walls a curve crosses (Theorem 2.2); at He criticality lengths are preserved and chords contract like  $3\pi/l$  (Theorem 2.4): the selected geometry folds. The same two-level structure reappears one level up: kinks are points with marks (kink density, covariance arms), and the third-cumulant memory is a sum of rank-one atoms over these points whose overlaps are codegrees in the covariance graph (Section 5).

**4. The observable and its estimators.** The mean is a boundary functional of the tiling: the *kink current*  $u = \sum_{l,a} \mathbb{E}[\delta(z_{l,a}) |\nabla z_{l,a}|^2 \partial_{h_l} h_L e_a]$  (Theorem 4.1), the first Schwinger–Dyson equation of the network’s Gaussian field theory (Section 7). A moment chain is an *unsharp* selection (gate projections replaced by their expectations), its third cumulant is a record of contacts (hub = kink, identity leg = pinned coordinate, Theorem 5.3), and its errors are read by a clock: Gaussian balls of uniform radius in the tiling, equivalently an exponentially distributed Ornstein–Uhlenbeck time, which turns Hermite contact order  $d$  into the Euler–Stein constant  $1/(d+2)$  (Theorems 6.1, 6.2).

| Sahasrabudhe [3]                          | Klartag [4]                          | Dirac ensembles<br>[8, 6, 5, 7]                 | He-ReLU network                              |
|-------------------------------------------|--------------------------------------|-------------------------------------------------|----------------------------------------------|
| Poisson points in a box                   | lattice points                       | random Dirac $D$                                | weight rows: random walls per layer          |
| conflict graph, codegree                  | contacts $\langle y, Ay \rangle = 1$ | spectral triple $(\mathcal{A}, \mathcal{H}, D)$ | tiling; codegree = arc-cosine kernel         |
| lift (forget embedding)                   | ellipsoid $A_t$ (metric)             | metric from $D$ (Connes)                        | selected Dirac $P_x D P_x$ ; pullback metric |
| nibble: $\text{Var} \leq p \sum c_y^2$    | rank-one contact updates             | action $\text{Tr } D^2, \text{Tr } D^4$         | He = quadratic action, critical gain         |
| small codegree $\Rightarrow$ independence | volume growth                        | 1-cut/2-cut, symmetry breaking                  | folding law; odd memory sector               |
| small-ball $\rho_h$ at scale $h$          | stopping when held                   | spectral dimension, variance                    | concentration coefficients at the kink       |

Two cautions recorded by the stage-9 workflow and kept here. The gate algebras are commutative subalgebras of  $L^\infty(\mathbb{R}^n, \gamma)$ , so the modular theory of their states is trivial; the noncommutative geometry of the problem lives in the pair (neuron algebra, Dirac/transport) and in the pair (neuron algebra, layer state on  $M_n$ ), Sections 3 and 5.4. And Klartag's process is the dual of a localization: there the metric is the martingale and the centre is fixed, here the centre is the martingale and the metric (precision) deterministic; the correspondence is between the contact structures, not the processes.

## 2 Two levels: random walls and their lift

### 2.1 Level one: the walls of a layer are an isotropic point process

Condition on layers  $1, \dots, l-1$ . The feature map  $x \mapsto h_{l-1}(x) \in \mathbb{R}^n$  is then fixed, and the rows  $w_{l,a}$  of  $W_l$  are  $n$  independent  $N(0, \frac{2}{n}I)$  vectors; their directions  $\hat{w}_{l,a}$  are  $n$  i.i.d. uniform points on  $S^{n-1}$ , a binomial process that is Poisson to first order. Each point carries a *wall* in input space,  $F_{l,a} = \{x : \langle w_{l,a}, h_{l-1}(x) \rangle = 0\}$ , and each input  $x$  carries a *cap*, the open hemisphere  $C_l(x) = \{u \in S^{n-1} : \langle u, h_{l-1}(x) \rangle > 0\}$ ; neuron  $a$  is active at  $x$  iff  $\hat{w}_{l,a} \in C_l(x)$ . This is the incidence structure of [3, §3.3] (points in a box, balls around them, intersections) with the rôles of the two families symmetric.

**Lemma 2.1** (Sign duality and the two codegrees). *Write  $\theta(u, v)$  for the angle between two vectors and  $J(\theta) = \sin \theta + (\pi - \theta) \cos \theta$ .*

- (i) (input side) *For inputs  $x, y$  with  $\theta = \theta(h_{l-1}(x), h_{l-1}(y))$ , conditionally on layers  $< l$  the codegree  $N_l(x, y) = \#\{a : a \text{ active at } x \text{ and at } y\}$  is  $\text{Bin}(n, \frac{\pi - \theta}{2\pi})$ , and the number of neurons whose gate differs at  $x$  and  $y$  is  $\text{Bin}(n, \frac{\theta}{\pi})$ . The weighted codegree  $K_l(x, y) = \langle h_l(x), h_l(y) \rangle$  has conditional mean  $\frac{1}{\pi} |h_{l-1}(x)| |h_{l-1}(y)| J(\theta)$  and conditional variance  $O(1/n)$  relative to its mean squared.*
- (ii) (neuron side) *At layer 1, for two rows with angle  $\theta_{ab}$ ,  $\gamma(a \text{ and } b \text{ active}) = \frac{\pi - \theta_{ab}}{2\pi}$  and  $\mathbb{E}_x[h_{1,a} h_{1,b}] = \frac{1}{2\pi} |w_a| |w_b| J(\theta_{ab})$ .*
- (iii) (Poisson covariance) *If the number of points is Poissonised to  $\text{Poi}(n)$ , the counts of points in two caps have  $\text{Cov}(N(C_l(x)), N(C_l(y))) = n \mu(C_l(x) \cap C_l(y)) = n \frac{\pi - \theta}{2\pi}$ : the overlap of two caps is the covariance of their counts, and the codegree is the pairing  $\tau(p_x p_y)$  of the two cap projections in  $L^\infty(S^{n-1}, \mu)$ .*

*Proof.* For a uniform direction  $u$  and two vectors at angle  $\theta$ ,  $\mathbb{P}(\langle u, v_1 \rangle > 0, \langle u, v_2 \rangle > 0) = \frac{\pi - \theta}{2\pi}$  (the lune between two hemispheres) and  $\mathbb{P}(\text{signs differ}) = \frac{\theta}{\pi}$ ; the rows are independent, which gives the binomial laws. The weighted codegree is a sum of  $n$  i.i.d. terms  $\text{relu}(\langle w, h(x) \rangle) \text{relu}(\langle w, h(y) \rangle)$ , whose mean is the degree-one arc-cosine kernel  $\frac{2}{n} \frac{|h(x)||h(y)|}{2\pi} J(\theta)$  [11]; the variance statement is the law of large numbers for  $n$  i.i.d. terms. (ii) is the same computation with  $x \sim \gamma$  isotropic and the rows fixed. (iii) is the Poisson covariance identity: split the two caps into the three disjoint pieces; counts of disjoint pieces are independent and the variance of a Poisson count equals its mean.  $\square$

So the post-activation covariance of a layer (the object every moment chain propagates) is the weighted codegree of the neurons' half-spaces, and the feature kernel on inputs is the weighted codegree of the inputs' caps. Neither side is privileged: the incidence  $(x, a) \mapsto \mathbf{1}[\langle w_a, h(x) \rangle > 0]$  is the sign pattern of one bilinear form.

## 2.2 Crofton: how many walls a curve crosses

**Theorem 2.2** (Crofton formula for the tiling). *Let  $c : [0, 1] \rightarrow \mathbb{R}^n$  be a piecewise  $C^1$  curve of inputs with  $h_{l-1}(c(t)) \neq 0$ , and let  $\Lambda_{l-1}(c) = \int_0^1 \left| \frac{d}{dt} h_{l-1}(c(t)) \right| dt$  be the angular length of its image in the layer- $(l-1)$  feature sphere. Conditionally on layers  $< l$ , the expected number of crossings of layer- $l$  walls by  $c$  is*

$$\mathbb{E}[\#\{(a, t) : z_{l,a}(c(t)) = 0\} \mid \text{layers} < l] = \frac{n}{\pi} \Lambda_{l-1}(c).$$

*Proof.* For one row,  $X_t = \langle w, \widehat{h_{l-1}(c(t))} \rangle \sqrt{n/2}$  is a centred Gaussian process with unit variance and  $\text{Cov}(X_t, \dot{X}_t) = 0$  (the derivative of a unit vector is orthogonal to it), so the Kac–Rice formula gives  $\mathbb{E}\#\{t : X_t = 0\} = \frac{1}{\pi} \int_0^1 \sqrt{\text{Var} \dot{X}_t} dt = \frac{1}{\pi} \int_0^1 \left| \frac{d}{dt} \widehat{h_{l-1}} \right| dt$ . The zeros of  $X_t$  are the crossings of the wall  $F_{l,a}$ ; sum over the  $n$  rows.  $\square$

This is the integral-geometric face of the two-level picture: the expected number of random walls (level one) that meet a set is the set's mean width in the geometry of the previous level. For a chord the count of *differing gates* is  $n\theta/\pi$  (Lemma 2.1); for the curve it is  $n\Lambda/\pi$ . The two differ as soon as the image curve folds.

## 2.3 The lift at He criticality: lengths are preserved, chords contract

**Proposition 2.3** (Unit angular gain). *Let  $v \perp c$  be a tangent vector at a feature vector  $c$ ,  $c' = \text{relu}(Wc)$ ,  $\dot{c}' = D(Wc)Wv$  its image,  $W$  with i.i.d.  $N(0, \sigma_w^2/n)$  entries. Then  $\mathbb{E}|c'|^2 = \frac{\sigma_w^2}{2}|c|^2$ ,  $\mathbb{E}|\dot{c}'|^2 = \frac{\sigma_w^2}{2}|v|^2$ ,  $\mathbb{E}\langle c', \dot{c}' \rangle = 0$  and  $\text{Var}\langle c', \dot{c}' \rangle = O(|c|^2|v|^2/n)$ . Hence the expected squared angular speed is multiplied by  $1 + O(1/n)$  for every  $\sigma_w$ , and lengths are preserved ( $\sigma_w^2 = 2$ ) iff the gated transfer has unit mean-square gain,  $\chi_1 = \sigma_w^2 \mathbb{E}[\text{relu}'(z)^2] = 1$ .*

*Proof.*  $\langle w_a, c \rangle$  and  $\langle w_a, v \rangle$  are independent Gaussians because  $v \perp c$ ; each neuron is active with probability  $\frac{1}{2}$  independently of  $\langle w_a, v \rangle$ . Then  $\mathbb{E}|c'|^2 = n \cdot \frac{1}{2} \cdot \frac{\sigma_w^2}{n} |v|^2$ ,  $\mathbb{E}|c'|^2 = n \cdot \mathbb{E} \text{relu}(g)^2$  with  $g \sim N(0, \sigma_w^2|c|^2/n)$ , and  $\langle c', \dot{c}' \rangle = \sum_a \text{relu}(\langle w_a, c \rangle) \langle w_a, v \rangle$  is a sum of  $n$  independent centred terms of variance  $\frac{\sigma_w^2|c|^2}{2n} \frac{\sigma_w^2|v|^2}{n}$ .  $\square$

**Theorem 2.4** (Folding law; infinite width). *In the limit  $n \rightarrow \infty$  with  $L$  fixed (the NNGP limit [11, 12]), the angle  $\theta_l$  between the features of two inputs obeys the  $\sigma_w$ -independent recursion  $\cos \theta_{l+1} = J(\theta_l)/\pi$ , and*

$$\theta_{l+1} = \theta_l - \frac{\theta_l^2}{3\pi} + O(\theta_l^3), \quad \theta_l = \frac{3\pi}{l + 3\pi/\theta_0 + O(\log l)}.$$

*The angular length of every curve is preserved,  $\Lambda_l = \Lambda_0$ . Consequently, for two inputs at angle  $\theta_0$ , the walls of layer  $l$  crossed by the great-circle arc between them number  $n\theta_0/\pi$  in expectation at every layer (in total  $Ln\theta_0/\pi$ ), while the gates that differ between the endpoints number  $n\theta_{l-1}/\pi \approx 3n/l$  (in total  $\approx 3n \log L$ ). The selected geometry is an isometric immersion of the input sphere into the feature sphere whose chords shrink like  $3\pi/l$ : it folds.*

*Proof.* The recursion is Lemma 2.1(i) with the law of large numbers, normalised by the norms (the  $\sigma_w$  factor cancels in the angle because  $\text{relu}$  is positively homogeneous). Expanding,  $\cos \theta' - \cos \theta = (\sin \theta - \theta \cos \theta)/\pi = \theta^3/(3\pi) + O(\theta^5)$ , and  $\cos \theta' = 1 - \theta'^2/2 + O(\theta'^4)$  give  $\theta'^2 = \theta^2 - \frac{2\theta^3}{3\pi} + O(\theta^4)$ , i.e.  $\theta' = \theta - \theta^2/(3\pi) + O(\theta^3)$ . Then  $1/\theta_{l+1} = 1/\theta_l + 1/(3\pi) + O(\theta_l)$ , and summing the  $O(\theta_l) = O(1/l)$  errors gives the  $O(\log l)$  term. The angle map  $f(\theta) = \arccos(J(\theta)/\pi)$  has  $f'(0) = 1$ , so the length of a curve,  $\lim \sum f(\Delta\theta_i) = \sum \Delta\theta_i$ , is preserved; this is the infinite-width form of Proposition 2.3. The crossing counts follow from Theorem 2.2 and Lemma 2.1(i).  $\square$

**Remark 2.5** (Two metrics on inputs). The *chord* metric  $|\hat{h}_l(x) - \hat{h}_l(y)| = 2\sin(\theta_l/2)$  and the *length* metric (the infimum of image lengths of curves joining  $x$  and  $y$ ) separate at depth: the first collapses like  $3\pi/l$ , the second is the round metric of the input sphere at every depth. Section 3.4 identifies the second as the Connes distance of the selected Dirac operator. Bias-free RELU networks are thus at the edge of chaos for every  $\sigma_w$  in the angle variable, and He initialisation is the point at which the length variable is critical too [12, 13]; the folding exponent is computed spectrally in Proposition 6.6. The linear-in- $L$  crossing count is the infinite-width form of the region counts of [14].

## 2.4 The lift as a tower of algebras

Let  $\mathcal{B}_l \subset L^\infty(\mathbb{R}^n, \gamma)$  be the von Neumann algebra generated by the gate functions  $\mathbf{1}[z_{k,a} > 0]$ ,  $k \leq l$ . Then  $\mathcal{B}_1 \subset \mathcal{B}_2 \subset \dots \subset \mathcal{B}_L$ ;  $\mathcal{B}_l$  is the algebra of functions constant on the tiles of the depth- $l$  tiling, its minimal projections are the tiles, and the trace  $\gamma$  gives each tile its Gaussian measure. The level-two data of layer  $l$  (the kernels  $K_l$  on inputs, the post-activation covariance  $C_l$  on neurons) are the ambient metric data of the level-one point process of layer  $l+1$ . The network is a  $\mathcal{B}_L$ -valued piecewise-linear section: on the tile with projection  $p_g$  it is the linear map  $J_g$ , so that  $h_L = \sum_g p_g J_g x$ . Everything that follows is about how the walls of this tower, and not its tiles, carry the mean.

## 3 The network as a random spectral triple

### 3.1 The chiral Dirac chain

**Definition 3.1** (Network spectral triple). Let  $X = \{0, \dots, L\} \times [n]$ ,  $\mathcal{A} = C(X) \cong \bigoplus_{l=0}^L \mathbb{C}^n$  acting diagonally on  $\mathcal{H} = \ell^2(X)$ , and

$$D = D_\downarrow + D_\downarrow^\top, \quad D_\downarrow = \sum_{l=1}^L W_l \otimes |l\rangle\langle l-1|,$$

with grading  $\Gamma = \bigoplus_l (-1)^l$  and real structure  $\mathcal{J} = \text{complex conjugation}$ . Then  $D\Gamma = -\Gamma D$ ,  $\mathcal{J}D = D\mathcal{J}$ ,  $\mathcal{J}\Gamma = \Gamma\mathcal{J}$ ,  $\mathcal{J}^2 = 1$ , and the order-zero and first-order conditions hold trivially because  $\mathcal{A}$  is commutative:  $(\mathcal{A}, \mathcal{H}, D, \Gamma, \mathcal{J})$  is a real even finite spectral triple of KO-dimension 0. Its commutators  $[D, f]$ ,  $f \in \mathcal{A}$ , have entries  $W_l(a, b) (f_{l-1, b} - f_{l, a})$ :  $D$  is the weighted adjacency of the layered graph, and the Connes distance of the triple is the metric on neurons in which strong weights are short edges.

**Proposition 3.2** (He initialisation as a critical Gaussian Dirac ensemble). *If the entries of every  $W_l$  are i.i.d.  $N(0, \sigma_w^2/n)$ , the law of  $D$  on the space of chiral chain operators is  $Z^{-1} \exp(-\frac{n}{4\sigma_w^2} \text{Tr } D^2) dD$ , the quadratic Barrett–Glaser action [8, 6] restricted to this kinematic space. For a fixed vector  $z$  and a gate projection  $P$  selecting the positive coordinates of  $z$ ,  $\mathbb{E}|W P z|^2 = \sigma_w^2 |P z|^2$  and  $\mathbb{E}|P z|^2 = \frac{1}{2}|z|^2$  for symmetric  $z$ ; the gated transfer has unit mean-square gain exactly at  $\sigma_w^2 = 2$ , and  $\mathbb{E} \log(|z_{l+1}|^2/|z_l|^2) = \log(\sigma_w^2/2) + O(1/n)$ . He initialisation is the critical coupling of the selected (gated) chain.*

*Proof.*  $\text{Tr } D^2 = 2 \sum_l \text{Tr } W_l^\top W_l$  and the Gaussian density of the entries is  $\prod \exp(-\frac{n}{2\sigma_w^2} W_{ij}^2)$ . For fixed  $y = Pz$ ,  $Wy$  has i.i.d.  $N(0, \sigma_w^2 |y|^2/n)$  coordinates. The logarithmic statement is the law of  $\log \chi_n^2/n$ , whose mean is  $-O(1/n)$ .  $\square$

### 3.2 The network is a compressed resolvent

**Theorem 3.3** (Gated resolvent). *For an input  $x$  let  $P_x \in \mathcal{A}$  be the projection that is the identity on layer 0 and the gate projection  $\mathbf{1}_{(0, \infty)}(z_l(x))$  on layer  $l \geq 1$ , and  $\psi = \bigoplus_l z_l$  the field of pre-activations ( $z_0 = x$ ). Then  $\psi$  is the unique solution of the self-consistent equation*

$$\psi = x \oplus 0 + D_\downarrow P_\psi \psi, \quad \text{i.e.} \quad \psi = (1 - D_\downarrow P_\psi)^{-1}(x \oplus 0),$$

where  $P_\psi = \mathbf{1}_{(0, \infty)}(\psi)$  is a spectral projection of the field in the commutative algebra. Since  $D_\downarrow P$  is nilpotent,  $(1 - D_\downarrow P)^{-1} = \sum_{k=0}^L (D_\downarrow P)^k$ , and

$$z_L(x) = [(1 - D_\downarrow P_x)^{-1}]_{L0} x = W_L P_{L-1} W_{L-1} \cdots P_1 W_1 x,$$

$$[z_L(x)]_k = \sum_{a_0 \rightarrow a_1 \rightarrow \cdots \rightarrow a_L = k} \prod_{l=1}^L W_l(a_l, a_{l-1}) \prod_{l=1}^{L-1} \mathbf{1}[z_{l, a_l}(x) > 0] x_{a_0}.$$

*Proof.* The equation is block lower triangular: its layer- $l$  component reads  $\psi_l = W_l P_{l-1} \psi_{l-1}$  with  $P_{l-1} \psi_{l-1} = \text{relu}(\psi_{l-1})$ , which is the forward pass; uniqueness follows by induction on  $l$ . The Neumann series terminates because  $(D_\downarrow P)^{L+1} = 0$ , and the  $(L, 0)$  block of  $(D_\downarrow P)^L$  is the displayed product, whose entries are the path sums.  $\square$

The network therefore computes with one fixed random operator  $D$  and one input-dependent projection  $P_x$  of the commutative algebra. Everything nonlinear is in the selection  $x \mapsto P_x$ ; given  $P_x$  the network is the resolvent of a compressed Dirac operator. The path expansion is the sum over walks of the counting problems of stage 9's reading (exterior-algebra path counts): the gates are input-dependent vertex weights.

### 3.3 The selected Dirac operator and the induced ensemble

**Definition 3.4** (Selection). The *selected Dirac operator* of  $x$  is  $D_x = P_x D P_x$ , the Dirac operator of the active subnetwork, and the *selected metric* is the pullback  $g_x = J(x)^\top J(x)$  of the output

metric,  $J(x) = \partial h_L / \partial x$ . The activation tiling is the partition of  $\mathbb{R}^n$  by the value of  $P_x$ ; the input law induces on the finite set of compressions  $\{P_g D P_g\}$  the probability  $\gamma(\text{tile } g)$ . We call this law the Dirac ensemble *induced by the input*: a random finite geometry drawn from the  $2^{n(L-1)}$  compressions of one random Dirac operator.

**Proposition 3.5** (Walls are rank-one contacts). *Let  $x$  cross the wall  $F_{l,a}$  transversally from  $z_{l,a} < 0$  to  $z_{l,a} > 0$ , all other gates fixed. Then  $P_x$  changes by the minimal projection  $e_{(l,a)}$ , the selected Dirac operator changes by  $e_{(l,a)} D P + P D e_{(l,a)} + e_{(l,a)} D e_{(l,a)}$  (a perturbation of rank at most two supported on the edges at the toggled site), the Jacobian changes by the rank-one operator*

$$\Delta J = (\partial_{h_l} h_L) e_a \nabla z_{l,a}^\top,$$

*and the selected metric changes by  $\Delta g = \nabla z_{l,a} v^\top J + J^\top v \nabla z_{l,a}^\top + |v|^2 \nabla z_{l,a} \nabla z_{l,a}^\top$  with  $v = (\partial_{h_l} h_L) e_a$ , a rank-two update whose new direction is the normal of the wall.*

*Proof.* Only the gate of  $(l, a)$  changes, so  $J = \partial_{h_l} h_L P_l \partial_x z_l$  changes by  $\partial_{h_l} h_L e_a e_a^\top \partial_x z_l$ , and  $e_a^\top \partial_x z_l = \nabla z_{l,a}^\top$ ; downstream gates are continuous across a generic wall. The other two statements are direct expansions.  $\square$

This is the precise sense in which the tiling plays the rôle of Klartag’s lattice. In [4] the ellipsoid matrix  $A_t$  moves until a lattice point  $y$  reaches its boundary, and the contact imposes the rank-one constraint  $\text{Tr}(A_t y y^\top) = 1$ . Here the selected geometry is constant inside a tile and receives a rank-one (Jacobian) or rank-two (metric) update at every wall a path crosses; along any path  $x_t$  the selected operators  $D_{x_t}$  form a Hamiltonian on a dynamically evolving graph whose edges switch at wall crossings. The localization processes of stage 9 are exactly such paths with a Gaussian ball attached: on Eldan’s path the ball  $N(m_t, (1+t)^{-1} I)$  shrinks around a martingale centre  $m_t$  until it lies in one tile, and the terminal selected operator is  $D_X$ ,  $X \sim \gamma$ . The process is dual to Klartag’s (there the metric is the martingale and the centre fixed); what transfers is the contact structure, and Section 6 shows that the estimator errors are read off exactly from it.

### 3.4 Distance on states

**Proposition 3.6** (Connes–Kantorovich distance of the selected geometry). *Let*

$$d_J(x, y) = \inf_c \int_0^1 |J(c(t)) \dot{c}(t)| dt$$

*over piecewise  $C^1$  curves  $c$  from  $x$  to  $y$  (the length of the image curve; a pseudometric because  $J$  may have a kernel), and for probability laws  $\nu, \nu'$  with first moments let*

$$d_D(\nu, \nu') = \sup\{|\nu(f) - \nu'(f)| : f \text{ 1-Lipschitz for } d_J\}.$$

*Then  $d_D = W_1^{d_J}$  (Kantorovich–Rubinstein), every output coordinate is 1-Lipschitz for  $d_J$ , and hence  $|u(\nu)_k - u(\nu')_k| \leq W_1^{d_J}(\nu, \nu')$  for the exact means of any two input laws. For a smooth nondegenerate metric  $d_D$  is Connes’ spectral distance with the Dirac operator of  $g$  [21, 20]; here it is its length-space extension to the piecewise-flat selected metric. In the infinite-width limit  $d_J$  restricted to the input sphere is the round metric at every depth (Theorem 2.4), while the chord metric of the features collapses: the distance on states is preserved by the selection and the distance between features is not.*

*Proof.* Kantorovich–Rubinstein duality on the (pseudo)metric space  $(\mathbb{R}^n, d_J)$ ;  $|h_{L,k}(x) - h_{L,k}(y)| \leq |h_L(x) - h_L(y)| \leq \int |J \dot{c}|$  for every curve. The last statement is Theorem 2.4.  $\square$

## 4 The mean is a kink current

### 4.1 The kink-current formula

Write  $\partial_{h_l} h_L(x)$  for the Jacobian of the output with respect to the post-activations of layer  $l$  ( $= I$  for  $l = L$ , and  $D_L W_L D_{L-1} \cdots W_{l+1}$  otherwise, with gates at  $x$ ), and  $f_{l,a}$  for the density of  $z_{l,a}(x)$ ,  $x \sim \gamma$ .

Call a tile *dead* if some hidden layer is entirely inactive on it ( $h_l \equiv 0$  there); on a dead tile the network vanishes identically and several walls coincide along its faces. The Gaussian measure of the dead tiles is  $\mathbb{P}(\exists l < L : h_l(x) = 0)$ , an orthant probability that is exponentially small in  $n$  ( $1.6 \cdot 10^{-1}$  at  $n = 6$ ,  $8 \cdot 10^{-6}$  at  $n = 16$ ,  $L = 3$ ; of order  $2^{-n}$  in general and below  $10^{-300}$  at  $n = 1024$ ).

**Theorem 4.1** (Kink current). *For almost every weight realisation, up to a remainder carried by the faces of the dead tiles,*

$$\begin{aligned} u = \mathbb{E} h_L(x) &= \sum_{l=1}^L \sum_{a=1}^n \mathbb{E} \left[ \delta(z_{l,a}(x)) |\nabla z_{l,a}(x)|^2 \partial_{h_l} h_L(x) e_a \right] \\ &= \sum_{l,a} f_{l,a}(0) \mathbb{E} \left[ |\nabla z_{l,a}|^2 \partial_{h_l} h_L e_a \mid z_{l,a} = 0 \right]. \end{aligned} \tag{1}$$

*Each term is the Gaussian surface integral of the wall  $F_{l,a}$ , weighted by the squared gradient of the pre-activation that defines it and by the transport of a unit kick at  $(l, a)$  to the output.*

*Proof.*  $F = h_L$  is piecewise linear and positively homogeneous, so Euler's relation  $F = x \cdot \nabla F$  holds off the walls, and  $\nabla F = J^\top$  is piecewise constant with jumps across the walls. Gaussian integration by parts for the bounded variation field  $J$  gives  $\mathbb{E} F = \mathbb{E}[x \cdot \nabla F] = \mathbb{E}[\text{div } \nabla F] = \mathbb{E}[\Delta F]$ , where  $\Delta F$  is the distributional Laplacian, a measure carried by the walls. Write  $J = D_L W_L D_{L-1} \cdots D_1 W_1$  and differentiate the gates:  $\partial_i D_l = \sum_a \delta(z_{l,a}) \partial_i z_{l,a} e_a e_a^\top$ . By the product rule

$$\sum_i (\partial_i J) \cdot i = \sum_{l,a} \delta(z_{l,a}) (\partial_{h_l} h_L) e_a \sum_i \partial_i z_{l,a} (e_a^\top W_l D_{l-1} \cdots W_1)_i = \sum_{l,a} \delta(z_{l,a}) |\nabla z_{l,a}|^2 (\partial_{h_l} h_L) e_a,$$

because  $e_a^\top W_l D_{l-1} \cdots W_1 = \nabla z_{l,a}^\top$ . Distinct walls meet in sets of codimension two for almost every weight realisation, the downstream gates are continuous across a generic point of  $F_{l,a}$ , and  $\{z_{l,a} = 0\}$  is a finite union of pieces of hyperplanes (on each live tile  $z_{l,a}$  is a nonzero linear form almost surely), so the products of distributions are well defined and  $\mathbb{E}[\delta(z)G] = f(0) \mathbb{E}[G \mid z = 0]$  with a continuous density  $f$ . Taking expectations gives (1). The only exception is a face of a dead tile: there a downstream pre-activation vanishes identically on one side, several gates jump at once, the product rule must be applied to the simultaneous jump, and  $z_{l,a}$  has an atom at 0; these faces carry the remainder, which is bounded by the Gaussian surface measure of the dead tiles' boundaries times the size of the transport. (The identity check confirms both halves: at  $n = 16$  the two sides agree to the Monte Carlo noise, 1.5%; at  $n = 6$ , where dead tiles carry 16% of the mass, they differ by 25%.)  $\square$

**Example 4.2.** For  $L = 1$ :  $f_{1,a}(0) = 1/(\sqrt{2\pi}|w_a|)$ ,  $|\nabla z_{1,a}|^2 = |w_a|^2$ , so  $u_a = |w_a|/\sqrt{2\pi}$ , the mean of  $\text{relu}(N(0, |w_a|^2))$ . For  $L = 2$  the layer-1 terms give  $D_2 W_2$  applied to the layer-1 kink masses on the walls, and the layer-2 terms give the kink masses of the curved walls  $\{w_{2,a} \cdot h_1(x) = 0\}$ .

**Theorem 4.3** (Tanaka form: the mean is transported local time). *Let  $B$  be a standard Brownian motion in  $\mathbb{R}^n$  started at 0 and  $L_t^{(l,a)}$  the semimartingale local time at 0 of  $z_{l,a}(B_t)$ . Then  $h_L(B_t)$  is a semimartingale with*

$$dh_L(B_t) = J(B_t) dB_t + \frac{1}{2} \sum_{l,a} (\partial_{h_l} h_L)(B_t) e_a dL_t^{(l,a)},$$

$$u = \mathbb{E} h_L(B_1) = \frac{1}{2} \sum_{l,a} \mathbb{E} \int_0^1 (\partial_{h_l} h_L)(B_t) e_a dL_t^{(l,a)}.$$

*In particular the kink mass  $m_{l,a} = \mathbb{E}[\delta(z_{l,a})|\nabla z_{l,a}|^2]$  is half the expected local time that  $z_{l,a}$  accumulates at its kink along a unit-time Brownian path from the origin.*

*Proof.*  $h_L$  is a difference of convex functions on each line, piecewise linear, and the Itô–Tanaka (Meyer) formula applies:  $dh_L(B) = \nabla h_L \cdot dB + \frac{1}{2} \Delta h_L(dt)$  with the Laplacian measure computed in the proof of Theorem 4.1; the occupation-time formula converts  $\delta(z_{l,a})|\nabla z_{l,a}|^2 dt$  into  $dL^{(l,a)}$ . Homogeneity gives  $\mathbb{E}[\delta(z(B_t))|\nabla z(B_t)|^2 G(B_t)] = t^{-1/2} \mathbb{E}[\delta(z)|\nabla z|^2 G]$  for degree-zero  $G$ , and  $\frac{1}{2} \int_0^1 t^{-1/2} dt = 1$ , consistent with (1).  $\square$

**Corollary 4.4** (Mean is created only at kinks). *For every neuron,  $\mathbb{E} h_{l,a} = m_{l,a} + \mathbb{E}[\mathbf{1}[z_{l,a} > 0] \Delta z_{l,a}]$ , where  $\Delta z_{l,a}$  is the wall measure of the lower layers transported to  $z_{l,a}$ . A layer adds mean exactly through its kink masses and transports the lower layers’ kink currents through its gates; a network without kinks has zero mean ( $\mathbb{E} z_l = \mathbb{E} \Delta z_l$  and  $\Delta$  of a linear map vanishes).*

*Proof.* Apply the computation of Theorem 4.1 to the single coordinate  $h_{l,a}$ :  $\Delta h_{l,a} = \delta(z_{l,a})|\nabla z_{l,a}|^2 + \mathbf{1}[z_{l,a} > 0] \Delta z_{l,a}$ .  $\square$

## 4.2 What an estimator has to compute: conditional expectations on walls

Formula (1) reduces the mean to two kinds of quantities per wall: the density  $f_{l,a}(0)$  of the pre-activation at its kink, and a conditional expectation *on the wall*, a measure-zero event. The first is the  $h \rightarrow 0$  limit of the concentration function  $Q_{l,a}(h) = \mathbb{P}(|z_{l,a}| \leq h) = 2f_{l,a}(0)h + \frac{1}{3}f_{l,a}''(0)h^3 + \dots$  (the pre-activation law is absolutely continuous with a continuous density, small-ball exponent one, no arithmetic structure: the gates are functions of a continuous Gaussian). On the official network 0 the Gaussian surrogate  $\varphi(\alpha)/\sigma$  of  $f_{l,a}(0)$  is right to  $1-3 \cdot 10^{-3}$  at every layer when the mean and variance are right (stage-9 workflow,  $2.6 \cdot 10^5$  inputs), so the densities are not where estimators fail. The second kind is a conditional expectation relative to an event of probability zero, the extreme case of the rare-event-relative conditioning of [3, §4] and [10]: the error tolerated in  $\mathbb{E}[\cdot | z_{l,a} = 0]$  is relative to the kink mass, not absolute. For a Gaussian state, conditioning on  $z_{l,a} = 0$  is a rank-one *pinning* (a directional localization along  $\nabla z_{l,a}$  taken to its end), and Section 5 shows that the third cumulant of every moment chain is the record of exactly these pinnings.

# 5 Estimators are unsharp selections; their memory is a record of contacts

## 5.1 Moment chains are unsharp selections

A moment chain does not know  $P_x$ ; it knows the Gaussian state  $N(m_l, S_l)$  of each layer and replaces the gate projection by its expectation, the diagonal effect  $\Phi_l = \text{diag } \Phi(\alpha_{l,a}) \in \mathcal{A}$ ,  $0 \leq \Phi_l \leq$

1. Its transport is  $T_{l \leftarrow l'} = W_l \Phi_{l-1} W_{l-1} \cdots \Phi_{l'} W_{l'+1}$ , the resolvent of the Dirac chain compressed by an effect instead of a projection.

**Proposition 5.1** (Unsharp selection). *With the path notation of Theorem 3.3,*

$$\mathbb{E}_x[\partial_{h_{l'}} z_l(x)] - T_{l \leftarrow l'} = \sum_{\text{paths}} \prod W \left( \mathbb{E}[\prod_k \mathbf{1}[z_{k,a_k} > 0]] - \prod_k \Phi_{k,a_k} \right),$$

*the sum over paths of the connected gate correlations along the path. The chain is exact for the expected transport iff the gates on every path are uncorrelated; its correction terms are correlations between the projection and the field that selects it (measurement back-action).*

*Proof.* Take expectations in the path expansion term by term.  $\square$

Only the on-wall transports of (1) enter the mean, so what matters is the back-action *conditional on a kink*. That is where the third cumulant comes from.

## 5.2 Pinning and births: the hub is a kink

**Lemma 5.2** (Pinning). *If  $z \sim N(m, S)$ , then  $z$  conditioned on  $z_a = 0$  is  $N(m - S e_a m_a / S_{aa}, S - S e_a e_a^\top S / S_{aa})$ : a rank-one update of the state along the covariance column of the pinned neuron. Downstream gate expectations shift to first order by  $-\varphi(\alpha_b) \rho_{ab} \alpha_a$ ,  $\rho_{ab} = S_{ab} / \sqrt{S_{aa} S_{bb}}$ .*

*Proof.* Gaussian conditioning;  $\Phi((m_b - S_{ab} m_a / S_{aa}) / \sigma_b) - \Phi(\alpha_b) = -\varphi(\alpha_b) \rho_{ab} \alpha_a + O(\rho^2)$ .  $\square$

**Theorem 5.3** (Births are contacts). *Let  $z \sim N(m, S)$  with small off-diagonal covariances and  $h = \text{relu}(z)$ . For distinct  $i, j, k$ ,*

$$\kappa_3(h_i, h_j, h_k) = \sum_{c \in \{i, j, k\}} \Phi(\alpha_a) S_{ac} \frac{\varphi(\alpha_c)}{\sigma_c} S_{cb} \Phi(\alpha_b) \Big|_{\{a, b\} = \{i, j, k\} \setminus \{c\}} + O(S_{\text{off}}^3),$$

*the sum over which of the three neurons is the hub. The hub contributes through  $\mathbb{E}[\text{relu}''(z_c)] = \varphi(\alpha_c) / \sigma_c = f_c(0)$ , its kink density, and the two arms are the covariances to the hub weighted by the gates. In CP form,  $\kappa_3 = \text{Sym} \sum_c (\Phi \circ S_c) \otimes 3e_c \otimes (f_c(0) S_c \circ \Phi)$  to second order in  $S_{\text{off}}$ , which is the star block of the organisers' chain  $(\Phi \circ S^{\text{off}}, 3I, (w_2 \circ S^{\text{off}} \circ \Phi)^\top)$ ,  $w_2 = f(0)$ ; its residual block makes the birth exact beyond second order.*

*Proof.* Joint cumulants of functions of a Gaussian vector expand over connected graphs on the index set with edges weighted by off-diagonal covariances and vertex factors  $\mathbb{E} f^{(\text{deg})}(z_v)$  (Price's theorem, or Wick's theorem applied to the Hermite expansions). On three vertices the connected graphs with two edges are the three paths, centred at the hub, with vertex factors  $\mathbb{E} \text{relu}'(z_a) = \Phi(\alpha_a)$ ,  $\mathbb{E} \text{relu}''(z_c) = \varphi(\alpha_c) / \sigma_c$ ,  $\mathbb{E} \text{relu}'(z_b) = \Phi(\alpha_b)$ ; the triangle is  $O(S_{\text{off}}^3)$ . Writing the hub as the middle leg  $e_c$  gives the CP form, and Sym distributes the three positions (factor 3).  $\square$

So the identity leg that makes the third-cumulant memory expensive is not an accident of a factorisation: it is the label of the kink at which the cumulant was born. Every birth block is  $n$  rank-one atoms, one per neuron  $c$  of the birth layer, with intensity  $f_c(0)$  and arms  $S_c$ ; transported, the atom of hub  $c$  born at  $l'$  and read at  $l$  is  $u_c z_c^\top$  with  $u_c = (T A e_c) \circ (T e_c)$  and  $z_c = T C e_c$ ,  $T = T_{l \leftarrow l'}$ . Lemma 5.2 says what the atom is: the first-order response of the downstream state to pinning neuron  $c$  at its kink. Measured on network 0 (stage-9 workflow): the atom energies are proportional to  $f_c(0)^2$  (log-log correlation 0.97–0.998 at ages 1, 3, 6), and the atoms are mutually orthogonal to  $10^{-3}$  ( $\|\sum_c P^{(c)}\|^2 / \sum_c \|P^{(c)}\|^2 = 0.998\text{--}1.004$ ).

### 5.3 Kinks as points: codegree and the frame bound

The second appearance of the two levels: the kinks of a layer are a marked point process on the neuron index (marks: density  $f_c(0)$ , arms  $S_c$ ), and the memory is the sum of their atoms. Two kinks overlap through the neurons their arms share,  $\langle S_c \circ \Phi, S_{c'} \circ \Phi \rangle = (S\Phi^2 S)_{cc'}$ : a codegree in the covariance graph. At He initialisation the off-diagonal covariances are  $O(n^{-1/2})$ , so codegrees are  $O(n^{-1/2})$  of the degrees, and the Gram matrix of the atoms, a Hadamard product of two such codegree matrices, has off-diagonal entries  $O(n^{-1})$  of the diagonal. This is exactly the small-codegree regime of [3, §3.3–3.4]. There it is what makes the nibble work (the conditional variance of a surviving degree is at most  $p \sum_y c_y^2$ ); here the same orthogonality makes the memory incompressible.

**Theorem 5.4** (Frame bound). *Let  $P = UZ^\top = \sum_{c=1}^N u_c z_c^\top$  with  $U, Z \in \mathbb{R}^{n \times N}$ , and let  $\hat{P} = UMZ^\top$  with rank  $M \leq k$  be any compression in the hub index (keeping  $k$  hubs, reweighting, or mixing hubs through any  $k$ -dimensional subspace). Then*

$$\|P - \hat{P}\|_F^2 \geq \lambda_{\min}(U^\top U) \|(I - M)Z^\top\|_F^2 \geq \lambda_{\min}(U^\top U) \lambda_{\min}(Z^\top Z) (N - k).$$

*Proof.*  $\|U(I - M)Z^\top\|_F^2 = \text{tr}(Z(I - M)^\top G_u(I - M)Z^\top) \geq \lambda_{\min}(G_u) \|(I - M)Z^\top\|_F^2$  with  $G_u = U^\top U$ , and  $\|(I - M)Z^\top\|_F^2 = \text{tr}((I - M)G_z(I - M)^\top) \geq \lambda_{\min}(G_z) \|I - M\|_F^2 \geq \lambda_{\min}(G_z)(N - k)$  by Eckart–Young.  $\square$

**Corollary 5.5** (The memory is as compressible as its transport leg, no more). *If the Hadamard leg is well conditioned,  $U^\top U \approx cI$  (orthogonal hub atoms of comparable size), every hub-index compression loses at least  $c$  times the energy of the transport leg  $Z = TC$  outside the kept subspace. The cheapest compression is therefore the low-rank truncation of  $TC$ , and its error is that truncation’s error. On network 0 the Hadamard bulk is near-isotropic while  $TC$  keeps 91/96/99% of the slice energy at rank 64/128/256 (stage-9 workflow), against the  $1 - 10^{-4}$  of energy that a  $10^{-2}$  relative slice accuracy requires; this is the measured failure of every tier in stage 9, now with its reason.*

### 5.4 Complementarity of the neuron algebra and the layer state

The gate algebras are commutative and the modular theory of their states is trivial. The genuinely noncommutative pair is the neuron algebra  $A = \text{diag}(\mathbb{R}^n) \subset M_n$ , where the nonlinearity acts (gates, Hadamard products, functional calculus), and the layer state  $\rho_l = M_l / \text{Tr } M_l$  on  $M_n$ ,  $M_l = \mathbb{E}[z_l z_l^\top]$ , faithful and non-tracial, whose modular group  $\sigma_s(Y) = \rho_l^{is} Y \rho_l^{-is}$  is the frame in which the transport acts.

**Proposition 5.6** (Takesaki for the neuron algebra). *The only conditional expectation of  $M_n$  onto  $A$  is the pinching  $\Delta(Y) = \text{diag}(Y)$ , and it preserves  $\rho$  iff  $\rho$  is diagonal; equivalently (Takesaki [19])  $A$  is  $\sigma^\rho$ -invariant iff  $\rho \in A$ . The independent-neuron (mean-field) coarse-graining is state-preserving only for uncorrelated layers.*

*Proof.* For a conditional expectation  $E$  onto  $A$  and  $i \neq j$ , bimodularity gives  $E(e_{ij}) = e_{ii}E(e_{ij})e_{jj} = 0$  since  $E(e_{ij})$  is diagonal; so  $E = \Delta$ .  $\text{Tr}(\rho \Delta Y) = \text{Tr}(\Delta(\rho)Y)$ , which equals  $\text{Tr}(\rho Y)$  for all  $Y$  iff  $\rho = \Delta(\rho)$ .  $\square$

**Theorem 5.7** (Complementarity). *Let  $B$  be the masa of the eigenprojections of  $\rho_l$  (simple spectrum),  $U$  its eigenbasis, and  $C_{ai} = |U_{ai}|^2 - \frac{1}{n}$ . With  $\tau = \frac{1}{n} \text{Tr}$ ,  $E_A = \Delta$  and  $E_B$  the pinching in*

the eigenbasis,  $\|E_B E_A - J\|_{L^2(\tau) \rightarrow L^2(\tau)} = \|C\|$ ,  $J(Y) = \tau(Y)1$ . Every  $\rho_l$ -preserving conditional expectation  $E_{\mathcal{N}}$  onto a subalgebra  $\mathcal{N} \subseteq B$  (any spectral coarse-graining of the layer state) factors through  $E_B$ , and for every  $Y \in A$  with  $\tau(Y) = 0$ ,

$$\|E_B Y\|_2 \leq \|C\| \|Y\|_2.$$

If the eigenvectors are Haar-distributed on the orthogonal group,  $\|C\| \approx 2\sqrt{2/n}$ .

*Proof.* The first identity is the alternating-expectation theorem for two partitions of unity into minimal projections (verified in the stage-9 workflow; it follows from writing  $E_A = VV^*$ ,  $E_B = WW^*$  with isometries onto the partitions with  $V^*W = C + \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ , so that  $E_B E_A - J = WC^\top V^*$  has norm  $\|C\|$ ). Subalgebras of  $B$  are invariant under  $\sigma^\rho$  because  $\rho \in B$ , so  $\rho$ -preserving expectations onto them exist; each is the identity on  $B$ -valued data after  $E_B$  (for  $\mathcal{N}$  spanned by blocks  $Q_k$  of eigenprojections,  $E_{\mathcal{N}}^\rho(Y) = \sum_k \text{Tr}(\rho Q_k Y) / \text{Tr}(\rho Q_k) Q_k$  and  $\text{Tr}(\rho Q_k Y) = \text{Tr}(\rho Q_k E_B(Y))$ ). For  $Y \in A$  with  $\tau(Y) = 0$ ,  $E_B Y = (E_B E_A - J)Y$ . For Haar orthogonal eigenvectors  $n|U_{ai}|^2$  is asymptotically  $\chi_1^2$ , so  $nC$  has centred entries of variance 2 and the norm of an  $n \times n$  such matrix is  $\approx 2\sqrt{2n}$ .  $\square$

At He initialisation the neuron algebra and the modular frame of the layer state are nearly mutually unbiased: a compression that respects the state (keeps eigen-directions of the covariance) sees a neuron-diagonal observable (a gate, a Hadamard product, a slice's hub diagonal) only to  $O(n^{-1/2})$  ( $2\sqrt{2/n} \approx 0.09$  at  $n = 1024$ ), and a compression that respects the neurons does not preserve the state. This is the Hadamard obstruction of stage 9 in operator-algebraic form, and it is why the spectral birth truncations of stage 9 saturate (rank 16 already keeps the spectral content and still loses a factor three): spectral information and readout information live in complementary masas.

## 5.5 Parity: the odd sector and broken symmetry

**Proposition 5.8** (Odd sector). *Fix layers  $1, \dots, l-1$ . The readout at layer  $l$  (before its gates) of any third-cumulant atom born at  $l' < l$  is a homogeneous cubic polynomial in  $W_l$ , hence odd under  $W_l \mapsto -W_l$ , and its conditional mean over the Gaussian law of  $W_l$  vanishes. Any surrogate that depends on  $W_l$  only through even statistics (spectra,  $W_l^\top W_l$ , the NNGP recursion, free-probability models) assigns it zero.*

*Proof.* Each of the three legs  $TAe_c$ ,  $Te_c$ ,  $TCE_c$  contains  $W_l$  exactly once as the last factor, and no gate follows it.  $\square$

In the language of [5]: the Gaussian weight ensemble has the symmetry  $W_l \mapsto -W_l$  for every  $l$ , the ensemble-averaged (free, NNGP) description is the symmetric phase, and each realisation carries an odd component of self-averaging size and realisation-specific sign, as the broken-symmetry equilibrium measures of the  $(1,0)$  ensemble do while the mean density of states stays symmetric. The analogy is exact in its consequence (ensemble surrogates cannot see the odd sector; the stage-9 workflow measured that its effect on the final mean is still half even, through odd readout coefficients, so surrogates capture at most a factor two of a memory worth a factor one hundred) and inexact in its mechanism: here the breaking is explicit at relative order  $n^{-1/2}$  (a random field), not spontaneous at order one. The collective scalars that the chain must track exactly—the radial fourth-cumulant scalar, worth a factor 16, the trace and the collective eigenvalue of the layer covariance—play the rôle of the multi-trace couplings ( $\text{tr } H^2$ )<sup>2</sup>,  $\text{tr } H$   $\text{tr } H^3$  of the Barrett–Glaser actions, whose values decide the phase there.

## 6 The clock of the selection: Mehler factors, modular time and the Euler–Stein constant

### 6.1 The Euler–Stein correction probes the estimator on balls of uniform radius

Stage 9 (Corollary *Eldan’s path*) gives, for a chain  $E$  exact on point masses and scale covariant, with  $e(y) = E(N(y, I))$  and  $\delta(y) = e(y) - y \cdot \nabla e(y) - \Delta e(y)$ ,  $u - e(0) = -\frac{1}{2} \int_0^\infty (1 + \tau)^{-3/2} \mathbb{E}_{y \sim N(0, \tau I)} \delta(y) d\tau$ .

**Theorem 6.1** (Uniform radius). *With  $r = (1 + \tau)^{-1/2} \in (0, 1]$ ,*

$$u - e(0) = - \int_0^1 \mathbb{E} \left[ \delta(M_r/r) \right] dr, \quad M_r \sim N(0, (1 - r^2)I).$$

*The localized laws on Eldan’s path are the Gaussian balls  $N(M_r, r^2 I)$  of radius  $r$  about a Gaussian centre, and under the defect weight their radius is uniformly distributed on  $[0, 1]$ ; equivalently the Ornstein–Uhlenbeck time  $t = -\log r$  of the localization is a standard exponential variable.*

*Proof.*  $d\tau = -2r^{-3}dr$ , so  $\frac{1}{2}(1 + \tau)^{-3/2}d\tau = \frac{1}{2}r^3 \cdot 2r^{-3}|dr| = dr$ . On Eldan’s path the localized law at time  $\tau$  is  $N(m, sI)$  with  $s = (1 + \tau)^{-1} = r^2$  and  $y = m/\sqrt{s} \sim N(0, \tau I)$ , i.e.  $m \sim N(0, r^2 \tau I) = N(0, (1 - r^2)I)$ . Finally  $\mathbb{P}(-\log r > t) = \mathbb{P}(r < e^{-t}) = e^{-t}$ .  $\square$

This is the precise form of “an ellipsoid in a tiling”: the correction is the average, over Gaussian balls of every radius  $r \in (0, 1)$  with equal weight, of the estimator’s defect on that ball. A ball of radius  $r$  meets a wall through its centre’s neighbourhood with probability proportional to  $r$  (Crofton), and the order at which the estimator mis-accounts the contact decides how fast its defect vanishes as the ball shrinks.

### 6.2 Hermite contact order and the Euler–Stein constant

**Theorem 6.2** (Contact order law). *Suppose the defect has the kink profile of order  $d$  at a wall through the origin,  $\delta(y) = c(\varphi \text{He}_d)(\langle v, y \rangle)$  with  $|v| = 1$  (the form of every layer-one wall of a bias-free network). Then  $\mathbb{E}_{y \sim N(0, \tau I)} \delta(y) = r^{d+1} \delta(0)$ , and the merge constant defined by  $u - e(0) = -a \delta(0)$  is*

$$a = \int_0^1 r^{d+1} dr = \frac{1}{d+2} = \mathbb{E}[e^{-(d+1)t}], \quad t \sim \text{Exp}(1).$$

*For odd  $d$  the defect vanishes at the origin and contributes nothing.*

*Proof.*  $\varphi \text{He}_d = (-1)^d \varphi^{(d)}$ , so  $\mathbb{E}_{s \sim N(0, \tau)} (\varphi \text{He}_d)(s) = (-1)^d (\varphi * \varphi_\tau)^{(d)}(0) = (-1)^d \varphi_{1+\tau}^{(d)}(0)$ , and  $\varphi_{1+\tau}(x) = r\varphi(rx)$  gives  $\varphi_{1+\tau}^{(d)}(0) = r^{d+1} \varphi^{(d)}(0) = (-1)^d r^{d+1} (\varphi \text{He}_d)(0)$ . Integrate against  $dr$  (Theorem 6.1).  $\square$

**Proposition 6.3** (Contact order is a concentration coefficient). *For a pre-activation with Gaussian surrogate  $N(m, \sigma^2)$ ,  $\alpha = m/\sigma$ , the concentration function at the kink is*

$$Q(h) = \mathbb{P}(|z| \leq h) = 2f(0)h + \frac{1}{3}f''(0)h^3 + O(h^5),$$

$$f(0) = \frac{\varphi(\alpha)}{\sigma} = \frac{(\varphi \text{He}_0)(\alpha)}{\sigma}, \quad f''(0) = \frac{(\varphi \text{He}_2)(\alpha)}{\sigma^3}.$$

The coefficient of  $h^{d+1}$  is the order- $d$  Hermite kink content. An estimator whose leading defect is an error in that coefficient has, by Theorem 6.2, merge constant  $1/(d+2)$ : the Euler–Stein constant reads off which Taylor coefficient of the concentration functions at the kinks the estimator gets wrong.

*Proof.*  $Q(h) = \int_{-h}^h f = 2f(0)h + \frac{1}{3}f''(0)h^3 + O(h^5)$ , and  $\partial_z^k [\varphi((z-m)/\sigma)/\sigma]_{z=0} = (-1)^k \sigma^{-k-1} \varphi^{(k)}(-\alpha)$ , with  $\varphi^{(k)} = (-1)^k \text{He}_k \varphi$  and parity.  $\square$

**Conjecture 6.4** (Contact order at depth). *For the chains of stage 9 the defect at depth is dominated by one contact order, and  $a = 1/(d+2)$  with:  $d = 0$  for the Gaussian closure and for chains with truncated third-cumulant memory (their errors are variance errors acting through  $f(0)$ ),  $d = 2$  for the exact first-order third-cumulant chain (its errors are second-order, the curvature  $f''(0)$ ).*

| chain (official networks, 2049 exact chain runs each)           | measured $a$  | predicted | order $d$ |
|-----------------------------------------------------------------|---------------|-----------|-----------|
| Gaussian closure, network 0 / 1                                 | 0.535 / 0.500 | 1/2       | 0         |
| window-one third-cumulant chain (memory dropped beyond one age) | 0.482         | 1/2       | 0         |
| exact first-order third-cumulant chain, network 0 / 1           | 0.255 / 0.257 | 1/4       | 2         |

The layer-wise profile of the absolute kink density along the localization was measured at all sixteen layers of network 0 to decay as  $(1 + \tau)^{-1/2} = r$  (stage-9 workflow), which is the  $d = 0$  profile; no profile of the exact chain’s defect has been measured, so the  $d = 2$  attribution rests on the constant. The earlier explanation of 0.255 as “two births, 1/3, reduced by rotation” is superseded: the order law gives 1/4 with no correction.

### 6.3 Mehler factors are imaginary modular times

**Proposition 6.5** (Mehler–KMS). *Let  $N$  be the Hermite number operator on  $L^2(\mathbb{R}, \gamma)$  ( $N\text{He}_k = k\text{He}_k$ ),  $\omega_\beta(Y) = \text{Tr}(\rho_\beta Y)$  with  $\rho_\beta = (1 - e^{-\beta})e^{-\beta N}$  the Gibbs state of  $N$ , faithful and non-tracial. For a standard Gaussian pair  $(X, Y)$  with correlation  $\rho = e^{-\beta} \in (0, 1)$  and real  $f, g \in L^2(\gamma)$ ,*

$$\mathbb{E} f(X)g(Y) = \langle f, e^{-\beta N} g \rangle = \frac{1}{1 - \rho} \omega_\beta(|g\rangle\langle f|).$$

The modular group of  $\omega_\beta$  is  $\sigma_s(Y) = e^{-i\beta s N} Y e^{i\beta s N}$ , the Mehler rotation (fractional Fourier transform of Gaussian space); the Ornstein–Uhlenbeck semigroup  $e^{-tN}$  is its continuation to imaginary time. The correlation of two Gaussian variables is thus the Boltzmann factor of one Hermite quantum at inverse temperature  $\beta = -\log \rho$ , and every pair correlation of the network, every Euler–Stein weight and every Wick coefficient of the chains is a matrix element of  $e^{-\beta N}$  at some  $\beta$ .

*Proof.* Mehler’s formula:  $\mathbb{E} f(X)g(Y) = \sum_k \rho^k \langle f, H_k \rangle \langle g, H_k \rangle$  with  $H_k = \text{He}_k / \sqrt{k!}$ ; and  $\text{Tr}(\rho_\beta |g\rangle\langle f|) = (1 - e^{-\beta}) \langle f, e^{-\beta N} g \rangle$ . The modular group of a Gibbs state  $\rho_\beta$  is  $\text{Ad } \rho_\beta^{is}$ .  $\square$

**Proposition 6.6** (The ReLU heat trace and the folding exponent). *Let  $K(\beta) = \mathbb{E}[\text{relu}(X) \text{relu}(Y)]$  at correlation  $e^{-\beta}$ , a weighted heat trace of the Hermite spectrum of  $\text{relu}$ :  $K(\beta) = \sum_k c_k^2 e^{-\beta k} / k!$  with  $c_0 = \frac{1}{\sqrt{2\pi}}$ ,  $c_1 = \frac{1}{2}$ ,  $c_{2j+2} = (-1)^j (2j - 1)!! / \sqrt{2\pi}$ ,  $c_{\text{odd} \geq 3} = 0$ . The weights decay like  $k^{-5/2}$  (the kink), and*

$$K(\beta) = \frac{1}{2} \left( 1 - \beta + \frac{2\sqrt{2}}{3\pi} \beta^{3/2} + O(\beta^2) \right), \quad \beta_{l+1} = \beta_l - \frac{2\sqrt{2}}{3\pi} \beta_l^{3/2} + O(\beta_l^2)$$

for the pair temperature  $\beta_l = -\log \cos \theta_l$  of two inputs. The linear coefficient is exactly the critical one (the depth flow is marginal at linear order because  $\text{relu}$  is homogeneous:  $K'(0)/K(0) = -\mathbb{E} \text{relu}^2 / \mathbb{E} \text{relu}^2 = -1$ ), and the kink's  $\beta^{3/2}$  term drives  $\beta_l \simeq 9\pi^2/(2l^2)$ , i.e.  $\theta_l \simeq 3\pi/l$ . Depth heats the neuron pairs quadratically,  $T_l = 1/\beta_l \simeq 2l^2/(9\pi^2)$ .

*Proof.*  $\mathbb{E}[\text{relu}(Z)\text{He}_k(Z)] = \mathbb{E}[\text{relu}^{(k)}(Z)] = \mathbb{E}[\delta^{(k-2)}(Z)] = \text{He}_{k-2}(0)\varphi(0)$  for  $k \geq 2$ , and  $\text{He}_{2j}(0) = (-1)^j(2j-1)!!$ ; Stirling gives  $c_{2j+2}^2/(2j+2)! \sim (8\pi^{3/2})^{-1}j^{-5/2}$ . In closed form  $K = \frac{1}{2\pi}(\sin \theta + (\pi - \theta) \cos \theta)$  with  $\cos \theta = e^{-\beta}$ , whose non-analytic part  $\frac{1}{2\pi}(\sin \theta - \theta \cos \theta) = \frac{\theta^3}{6\pi} + O(\theta^5)$  with  $\theta = \sqrt{2\beta}(1 + O(\beta))$  is  $\frac{\sqrt{2}}{3\pi}\beta^{3/2}$ . The recursion is  $e^{-\beta_{l+1}} = K(\beta_l)/K(0)$ , and  $\beta^{-1/2}$  grows by  $\frac{\sqrt{2}}{3\pi}$  per layer.  $\square$

**Remark 6.7** (One clock). Three flows are one structure. Depth raises the pair temperature,  $\beta_l \rightarrow 0$  like  $l^{-2}$ , with exponent fixed by the Weyl-type tail  $k^{-5/2}$  of the activation's Hermite spectrum (a smooth activation has an analytic heat trace and flows exponentially unless tuned). Localization lowers it: the ball radius is the Boltzmann factor  $r = e^{-t}$  of the Ornstein–Uhlenbeck time, and the Euler–Stein constant  $1/(d+2) = \mathbb{E}e^{-(d+1)t}$  is the mean Boltzmann factor of  $d+1$  quanta at an exponential time. The quantum Hermite transform of stage 9's reading is the real-time version of the same group. In this sense the Gaussian input's modular flow, the network's depth flow and the estimator's error clock are the imaginary-time, discrete-time and averaged-time faces of the single semigroup  $e^{-tN}$ .

## 7 Bootstrap: the mean as the unique solution of a positivity problem

Khalkhali and Pagliaroli [6] solve Dirac ensembles without solving them: Schwinger–Dyson (loop) equations give linear relations among moments, positivity of moment matrices cuts out a convex island, and the island shrinks onto the answer as the moment degree grows. The network admits the same treatment, with the Gaussian input as the “action”, Stein's identity as the loop equation, and the gates as idempotent variables.

**Definition 7.1** (The ReLU moment problem). Let  $\mathbb{R}[x, \theta]$  be the polynomial algebra in  $x \in \mathbb{R}^n$  and commuting idempotents  $\theta_{l,a}$  ( $\theta^2 = \theta$ ),  $l = 1, \dots, L$ . Define recursively the polynomials  $z_1 = W_1x$ ,  $z_{l,a} = \sum_b W_l(a, b) \theta_{l-1,b} z_{l-1,b}$ . A *feasible moment functional* is a linear  $\mathcal{L} : \mathbb{R}[x, \theta] \rightarrow \mathbb{R}$  with  $\mathcal{L}(1) = 1$ , Gaussian  $x$ -moments (equivalently the loop equations  $\mathcal{L}(x_i q) = \mathcal{L}(\partial_i q)$  for  $q \in \mathbb{R}[x]$ ), positive moment matrix  $\mathcal{L}(p^2) \geq 0$ , and positive localizing forms  $\mathcal{L}(p^2 \theta_{l,a} z_{l,a}) \geq 0$ ,  $\mathcal{L}(-p^2(1 - \theta_{l,a})z_{l,a}) \geq 0$  for all  $p$ .

**Theorem 7.2** (The island is a point). *For almost every weight realisation every feasible moment functional represented by a measure is the law of  $(x, \theta(x))$  except for the values of the gates on dead tiles, where all downstream pre-activations vanish and the gates are unconstrained; these measures are moment-determinate, and they all give the same value to every pre-activation moment. In particular  $u = \mathcal{L}(\theta_L \circ z_L)$  is determined by positivity and the loop equations alone.*

*Proof.* A representing measure  $\mu$  lives on  $\mathbb{R}^n \times \{0, 1\}^{nL}$  and is supported in  $K = \{\theta_{l,a} z_{l,a} \geq 0, (1 - \theta_{l,a}) z_{l,a} \leq 0\}$ . If  $z_{1,a}(x) > 0$  the second constraint forces  $\theta_{1,a} = 1$ , if  $z_{1,a}(x) < 0$  the first forces  $\theta_{1,a} = 0$ ; given the gates of layers  $< l$ ,  $z_l(x, \theta)$  is the true pre-activation and the same argument forces layer  $l$ . So the fibre of  $K$  over  $x$  is the single point  $\theta(x)$  unless some  $z_{l,a}(x) = 0$ . On a live tile this happens on a  $\gamma$ -null set ( $z_{l,a}$  is a nonzero linear form there almost surely); on a

dead tile, where  $h_l \equiv 0$ , all later pre-activations vanish whatever the later gates are, so the fibre is a set of gate patterns on which every  $z_{l'}, l' > l$ , and  $h_L$  take the value 0. Since the  $x$ -marginal is  $\gamma$ ,  $\mu$  is the graph measure up to the choice of these irrelevant gates. Determinacy:  $\mu = \sum_{\vartheta} \mu_{\vartheta} \otimes \delta_{\vartheta}$  over the finitely many gate patterns, with  $\mu_{\vartheta} \leq \gamma$ ; the moments of  $\mu$  determine those of every  $\mu_{\vartheta}$  (the monomials in  $\theta$  span the indicators of patterns), and each  $\mu_{\vartheta}$  satisfies Carleman's condition because its moments are bounded by the Gaussian ones.  $\square$

Under the standard convergence theory of moment–sum-of-squares relaxations for determinate measures [18], the islands of degree  $2k$  form a nested sequence of intervals containing  $u$  and shrinking to it. Two features are specific to the network.

**Proposition 7.3** (Loop equations need the kink generators). *Stein's identity for test functions that involve gates produces the distributions  $\delta(z_{l,a})$ :  $\partial_i \theta_{l,a} = \delta(z_{l,a}) \partial_i z_{l,a}$ . The loop equations therefore close only on the algebra generated by  $x$ , the gates and the kink distributions  $\delta^{(k)}(z_{l,a})$ . The first of them, for the test field  $q = J_{\cdot i}$ , is the kink-current formula (1); the Wick coefficients  $\mathbb{E}[\partial^k \text{relu}^p]$  of every cumulant chain are the Gaussian-state values of these kink generators.*

*Proof.* Differentiate  $\theta_{l,a}(x) = \mathbf{1}[z_{l,a}(x) > 0]$  in the sense of distributions; the first loop equation is  $\mathbb{E}[x \cdot \nabla h_L] = \mathbb{E}[\Delta h_L]$ , computed in Theorem 4.1. For a Gaussian  $z$ ,  $\mathbb{E}[\delta^{(k)}(z)]$  are the derivatives of the density at the kink, which is what the chains' Wick tables contain.  $\square$

**Proposition 7.4** (The level-one island of a single neuron is Scarf's interval). *If only the mean  $m$  and variance  $s^2$  of  $z_{l,a}$  are imposed, together with  $h \geq 0$ ,  $h \geq z$ ,  $h(h - z) = 0$ , the closure of the set of feasible values of  $\mathbb{E}h$  is exactly  $[\max(0, m), \frac{1}{2}(m + \sqrt{m^2 + s^2})]$ . The Gaussian closure picks the interior point  $s(\alpha\Phi(\alpha) + \varphi(\alpha))$ ,  $\alpha = m/s$ .*

*Proof.* The constraints force  $h = z_+$ ; the lower bound is Jensen, and the upper bound is Scarf's worst case over laws with given mean and variance, attained by a two-point law [15].  $\square$

So a moment chain is a truncated Schwinger–Dyson hierarchy solved with a Gaussian (or Edgeworth) ansatz and *without* positivity. Positivity is what would turn a chain into a certificate; its cost at width 1024 is prohibitive at any useful degree, and its role here is structural: it shows that the exact answer is the unique point of an island cut out by the input's loop equations and the complementarity of the gates, and that the information a chain discards is the positivity of the kink generators' moment matrices, which is exactly what goes wrong when a closure produces a negative fourth cumulant or a variance outside its admissible range.

## 8 The token picture: vertices, quadratic forms, two-token walks

This section develops the simplest framing of the problem: every input slot is a vertex, the weights are weighted edges, a layer is a quadratic form on the vertex graph, the next layer is read from the two-token (Kikuchi) lift of that graph, and a second tier—the input law—sets the environment in which the lower tier's walks take place. The framing is exact: it is the Wiener chaos (bosonic Fock) structure of Gaussian space, with tokens as Hermite quanta. It reorganises everything above in a few lines, and it locates the obstruction precisely: it is a contact of tokens at a hidden neuron.

## 8.1 Dictionary

Let  $V = [n]$  be the input slots. By the Wiener–Itô isomorphism

$$L^2(\mathbb{R}^n, \gamma) \cong \bigoplus_{k \geq 0} \text{Sym}^k(\ell^2(V)), \quad F = \sum_k \langle f_k, :x^{\otimes k}: \rangle, \quad f_k = \frac{1}{k!} \mathbb{E}[\nabla^k F] \quad (\text{Stroock}),$$

the  $k$ -th chaos is spanned by the multivariate Hermite polynomials  $\text{He}_\alpha$ ,  $|\alpha| = k$ : configurations of  $k$  indistinguishable tokens on the vertices, repetitions allowed (the bosonic  $k$ -token Kikuchi level). The Gaussian input is the vacuum  $\Omega = 1$ ;  $a_i = \partial_i$  removes a token at  $i$ ,  $a_i^\dagger = x_i - \partial_i$  adds one,  $N = \sum_i a_i^\dagger a_i$  is the Ornstein–Uhlenbeck number operator. A weighted graph on  $V$  is a function on the pair groupoid  $V \times V$ , i.e. an element of  $M_n = \mathbb{C}[V \times V]$ ; it moves one token along an edge as  $d\Gamma(A)$  and all tokens as  $\Gamma(A)$ . Two-token kernels are quadratic forms, i.e. elements of the groupoid algebra, and contracting two kernels in  $r$  shared token slots,  $f \otimes_r g$ , is groupoid convolution ( $Q \otimes_1 Q' = QQ'$ ,  $Q \otimes_2 Q' = \text{Tr } QQ'$ ). Positive definite functions on the pair groupoid act by Schur multiplication as completely positive maps; the unit space (the diagonal) is where two tokens sit on one vertex.

## 8.2 Layer one: each neuron is a one-mode token tower, and its mean is a trace

**Proposition 8.1** (Ridge towers). *With  $c_k = \mathbb{E}[\text{relu}(Z)\text{He}_k(Z)]$ , i.e.  $(c_0, c_1, c_2, c_3, c_4, c_5, c_6, \dots) = (\varphi_0, \frac{1}{2}, \varphi_0, 0, -\varphi_0, 0, 3\varphi_0, \dots)$ ,  $\varphi_0 = 1/\sqrt{2\pi}$ ,  $c_k = \text{He}_{k-2}(0)\varphi_0$  for  $k \geq 2$ ,*

$$h_{1,a} = |w_a| \sum_k \frac{c_k}{k!} (a^\dagger(\hat{w}_a))^k \Omega, \quad f_2(h_{1,a}) = Q_a = \frac{w_a w_a^\top}{2\sqrt{2\pi} |w_a|}.$$

*Each layer-one neuron lives in the one-mode Fock space of its own edge vector, and its quadratic form is the rank-one graph of that edge. Its degree- $k$  coefficient is  $\mathbb{E}[\text{relu}^{(k)}]$ : the gate for  $k = 1$ , the kink density for  $k = 2$ , and derivatives of the kink for  $k \geq 3$ .*

*Proof.*  $\text{relu}(|w|s) = |w| \text{relu}(s)$  with  $s = \hat{w} \cdot x \sim N(0, 1)$ , the one-dimensional Hermite expansion, and  $\text{He}_k(\hat{w} \cdot x) = (a^\dagger(\hat{w}))^k \Omega$ .  $c_k = \mathbb{E}[\text{relu}^{(k)}] = \mathbb{E}[\delta^{(k-2)}] = \text{He}_{k-2}(0)\varphi_0$  by Gaussian integration by parts.  $\square$

**Theorem 8.2** (Euler–Fock ladder). *If  $F$  is positively homogeneous of degree one then  $(N + \sum_i a_i a_i)F = F$ , i.e.*

$$f_0 = 2 \text{tr } f_2, \quad \text{tr}_{23} f_3 = 0, \quad f_k = -\frac{(k+1)(k+2)}{k-1} \text{tr}_{k+1, k+2} f_{k+2} \quad (k \geq 2).$$

*In particular the mean of every neuron of every layer is twice the trace of its quadratic form: the mean is a closed two-token loop, the two tokens of the quadratic form placed on the same input vertex and summed.*

*Proof.*  $x \cdot \nabla = \sum_i (a_i^\dagger + a_i) a_i = N + \sum_i a_i a_i$ , and Euler’s relation  $x \cdot \nabla F = F$  holds for one-homogeneous  $F$ .  $\sum_i a_i a_i$  maps the chaos- $(k+2)$  component with kernel  $f_{k+2}$  to chaos  $k$  with kernel  $(k+2)(k+1) \text{tr } f_{k+2}$ , the trace over two token slots. Compare the chaos- $k$  components. For  $h_{1,a}$ :  $2 \text{tr } Q_a = |w_a|/\sqrt{2\pi}$ , and  $-12 \text{tr } f_4 = f_2$  with  $f_4 = -\frac{\varphi_0}{24} |w_a| \hat{w}_a^{\otimes 4}$ .  $\square$

### 8.3 Layer two: the quadratic form is the graph of two-step walks

**Theorem 8.3** (Layer-two tokens). For  $z_{2,c} = \sum_a W_2(c, a) h_{1,a}$ , with  $\omega = W_2(c, \cdot)^\top$ ,

$$z_{2,c} = m_c + \langle f_c, x \rangle + :x^\top Q_c x: + (\text{chaos} \geq 4),$$

$$f_c = \frac{1}{2} W_1^\top \omega, \quad Q_c = W_1^\top D_c W_1, \quad D_c = \text{diag} \left( \frac{W_2(c, a)}{2\sqrt{2\pi}|w_a|} \right).$$

$Q_c(i, j) = \sum_a W_1(a, i) D_c(a) W_1(a, j)$  is the weighted graph of two-step walks  $i \rightarrow a \rightarrow j$  through the hidden layer, turning at  $a$  with weight  $D_c(a)$ . By Sylvester's identity every spectral quantity of  $Q_c$  is one of  $D_c G$ , where  $G = W_1 W_1^\top$  is the codegree graph of the hidden neurons (overlaps of their input supports; normalised, the cosines  $\rho_{ab}$ ).

*Proof.* Linearity of the chaos decomposition and Proposition 8.1 ( $\mathbb{E} g_{1,a} = \frac{1}{2}$ ).  $\square$

**Theorem 8.4** (Cumulants are walk sums). (i) For  $z = m + \langle f, x \rangle + x^\top Q x - \text{tr } Q$ ,

$$\log \mathbb{E} e^{itz} = it(m - \text{tr } Q) - \frac{1}{2} \log \det(I - 2itQ) - \frac{t^2}{2} f^\top (I - 2itQ)^{-1} f,$$

$$\begin{aligned} \kappa_r(z) &= 2^{r-1}(r-1)! \text{Tr } Q^r + r! 2^{r-3} f^\top Q^{r-2} f \\ &= 2^{r-1}(r-1)! \text{Tr}(D_c G)^r + r! 2^{r-5} \omega^\top G(D_c G)^{r-2} G \omega : \end{aligned}$$

closed walks on the hidden codegree graph, and open walks whose two ends are the linear token. (ii) Exactly (all chaoses),  $\kappa_r(z_{2,c}) = \sum_{a_1, \dots, a_r} \prod_j \omega_{a_j} |w_{a_j}| \kappa(\text{relu}(s_{a_1}), \dots, \text{relu}(s_{a_r}))$ , and for distinct indices the joint cumulant of the ridge functions is the sum over connected multigraphs on  $\{a_1, \dots, a_r\}$  with weight  $\rho^m/m!$  on an edge of multiplicity  $m$  and weight  $c_{\deg(v)}$  on a vertex of degree  $\deg(v)$  (the diagram formula). For  $r = 3$  the leading graphs are the three paths, centred at a hub of degree two (a kink,  $c_2 = \varphi_0$ ) with two gated arms ( $c_1 = \frac{1}{2}$ ), and the triangle ( $c_2^3$ ). Repeated indices (two or three tokens on the same hidden neuron: contacts) are given by the exact single-neuron and pair laws, not by the quadratic truncation.

*Proof.* (i) Diagonalise  $Q$ ; each mode is a shifted noncentral  $\chi^2$ , and completing the square gives the displayed log-characteristic function. Expand  $\log \det(I - 2itQ) = -\sum_r (2it)^r \text{Tr } Q^r / r$  and  $(I - 2itQ)^{-1} = \sum_s (2it)^s Q^s$ , and read off  $r!$  times the coefficient of  $(it)^r$ . Substitute  $Q = W_1^\top D_c W_1$  and  $f = \frac{1}{2} W_1^\top \omega$ . (ii) Mehler's formula for Gaussian vectors with unit variances:  $\mathbb{E} \prod_j g_j(s_j) = \sum \prod_v \mathbb{E}[g_v^{(\deg v)}] \prod_e \rho_e^{m_e}/m_e!$  over multigraphs; cumulants keep the connected ones.  $\square$

| width | $\kappa_3(z_{2,c})$ (MC) | self   | contacts | distinct neurons | leading diagrams |
|-------|--------------------------|--------|----------|------------------|------------------|
| 24    | -0.521                   | -0.465 | -0.115   | 0.059            | 0.064            |
| 96    | 0.210                    | 0.078  | 0.121    | 0.0115           | 0.0133           |

(Self: all three indices on one neuron; contacts: two of three on one neuron, by the exact pair law; distinct: Monte Carlo minus the two exact parts; leading diagrams: hub paths and triangles.) The distinct-neuron part is the walk sum, to the accuracy of its leading diagrams; at these widths the contacts carry most of a single pre-activation's third cumulant. The quadratic truncation of (i) applied to the whole of  $z_{2,c}$  gives 0.070 against the true -0.521 ( $n = 24$ ): the walk picture is right for exchange between distinct neurons and wrong for contacts, which must be computed exactly.

## 8.4 How many tokens: depth heats the tower

**Proposition 8.5** (Mean token number). *For  $F \in L^2(\gamma)$  the mean token number of its centred part is  $\bar{N}(F) = \mathbb{E}|\nabla F|^2 / \text{Var } F$  (since  $\mathbb{E}|\nabla F|^2 = \sum_k k k! \|f_k\|^2$  and  $\text{Var } F = \sum_{k \geq 1} k! \|f_k\|^2$ ). In the infinite-width limit the pre-activations of layer  $l$  have*

$$\bar{N}_l = \frac{1}{1 - \cos \theta_{l-1}}, \quad \theta_0 = \frac{\pi}{2}, \quad \cos \theta_{k+1} = J(\theta_k)/\pi, \quad \bar{N}_l \simeq \frac{2l^2}{9\pi^2} \quad (l \rightarrow \infty),$$

the pair temperature of Proposition 6.6.

*Proof.* For a one-homogeneous random field with kernel  $K(x, y) = \frac{|x||y|}{n} k(\hat{x} \cdot \hat{y})$  differentiating at  $y = x$ ,  $|x|^2 = n$ , gives  $\mathbb{E}|\nabla_x z|^2 = \sum_i \partial_{x_i} \partial_{y_i} K|_{y=x} = \frac{1}{n}(k(1) + (n-1)k'(1))$  while  $\mathbb{E}z^2 = k(1)$ ; the normalised dual kernel of RELU has derivative 1 at  $t = 1$  at every layer (criticality), so  $\mathbb{E}|\nabla z|^2 = \mathbb{E}z^2(1 + O(1/n))$ . The squared mean  $m^2$  is the kernel at two independent inputs, whose cosine tends to 0, so  $m^2/\mathbb{E}z^2 \rightarrow \cos \theta_{l-1}$  with the angle recursion of Theorem 2.4 started at  $\pi/2$ . Then  $\bar{N} = \mathbb{E}z^2/(\mathbb{E}z^2 - m^2)$ ; the asymptotics follow from  $\beta_l \simeq 9\pi^2/(2l^2)$ .  $\square$

| layer                   | 1    | 2    | 3    | 5    | 9    | 13   | 16    |
|-------------------------|------|------|------|------|------|------|-------|
| infinite width          | 1.00 | 1.47 | 1.98 | 3.13 | 6.04 | 9.72 | 12.99 |
| width 256 (Monte Carlo) | 1.00 | 1.47 | 2.01 | 3.12 | 6.26 | 9.46 | 10.56 |

So the token tower does not truncate: the last pre-activation of a sixteen-layer network carries eleven to thirteen tokens on average, far from the two of a quadratic form (and far below the  $2^{l-1}$  of naive degree doubling). Its marginal law is nevertheless close to Gaussian (skewness about 0.1 on network 0). The two facts are reconciled by the fourth-moment theorem of Nualart and Peccati [16]: within a chaos, Gaussianity is controlled by the contractions  $f \otimes_r f$  ( $1 \leq r < k$ ), the overlaps of a kernel's token configurations with themselves, i.e. token codegrees, and not by the token number. The Gaussian closure works because these contractions are small at He initialisation, and it fails exactly where they are not: at contacts, and along the collective mode.

## 8.5 Two tiers: the input law is the environment

**Theorem 8.6** (One-token plus two-token). *For a one-homogeneous  $F$  and every Gaussian input law  $N(m, \Sigma)$ ,*

$$\mathbb{E}_{N(m, \Sigma)} F = \langle m, \mathbb{E} \nabla F \rangle + \langle \Sigma, \mathbb{E} \nabla^2 F \rangle,$$

with the distributional Hessian (walls). Equivalently, the exact mean  $u(m, \Sigma)$  satisfies Euler's relation and the heat equation  $\partial_{\Sigma_{ij}} u = \frac{1}{2} \partial_{m_i} \partial_{m_j} u$ ; in token language, adding a token pair at  $(i, j)$  to the input state has the same effect as two successive displacements along  $i$  and  $j$ .

*Proof.*  $\mathbb{E}F = \mathbb{E}[x \cdot \nabla F] = \langle m, \mathbb{E} \nabla F \rangle + \mathbb{E}[(x - m) \cdot \nabla F]$  and Stein's identity for  $N(m, \Sigma)$ ,  $\mathbb{E}[(x - m)g] = \Sigma \mathbb{E} \nabla g$ , with  $g = \nabla F$ . The heat equation is  $\partial_{\Sigma} \mathbb{E}F(m + \Sigma^{1/2}Z) = \frac{1}{2} \mathbb{E} \nabla^2 F$ .  $\square$

The upper tier is the input law; it sets the environment of the lower tier, namely the kink densities and the gate-pair kernels of every layer. A moment chain  $e(m, \Sigma)$  obeys Euler's relation (scale covariance) but not the heat equation: its pair response and its double-displacement response disagree, and that disagreement is the heat defect of stage 9. Along a localization (Eldan's balls of radius  $r$  about a moving centre, Theorem 6.1) the environment cools. Gates freeze, the contact weights  $\Phi(1 - \Phi)$  vanish away from walls, the two-token term transfers into the one-token term, and at  $r = 0$  the walk is free and every chain is exact. The Euler–Stein correction integrates the chain's mismatch over this cooling, and its constant  $1/(d+2)$  records the token order of the mismatch at the kink.

## 8.6 The two-token walk and its environment

Pathwise, the gradient Gram  $\Gamma_l = J_l J_l^\top$  ( $J_l = \partial z_l / \partial x$ ) obeys  $\Gamma_l = W_l D_{l-1} \Gamma_{l-1} D_{l-1} W_l^\top$ : two tokens walk up the layered graph, each picking up weights and gates. In a factorised environment its expectation is the completely positive map  $\Gamma \mapsto W(K \circ \Gamma)W^\top$ , where the gate-pair kernel  $K(a, b) = \mathbb{P}(z_a > 0, z_b > 0)$  is a positive definite function on the pair groupoid. The tetrachoric (Mehler) series splits it into three interactions:

$$K = \underbrace{\Phi \Phi^\top}_{\text{independent walkers}} + \underbrace{\text{diag}(\Phi - \Phi^2)}_{\text{contact}} + \underbrace{\sum_{k \geq 1} \frac{1}{k!} (\rho^{\circ k} \circ \psi_k \psi_k^\top)_{\text{off}}}_{\text{exchange of } k \text{ quanta}}, \quad \psi_k = \varphi(\alpha) \text{He}_{k-1}(\alpha).$$

The first term is a congruence (the two tokens move independently with mean gates). The second acts only when both tokens sit on one neuron, and it is diagonal, so it is sparse in the two-token graph. The third is dense, with off-diagonal entries of order  $n^{-k/2}$ . The mean reads the unit space of the final quadratic form (its trace).

**Proposition 8.7** (Independent walkers are wrong at order one, by exactly the contact term). *For two layers, let  $u_c^{\text{walk}} = \Phi_c m_c + f_c(0) \mathbb{E}|\nabla z_{2,c}|^2$  be the kink current with independent walkers, and  $u_c^{\text{GC}} = \Phi_c m_c + f_c(0) \sigma_c^2$  the Gaussian closure (same exact  $m_c, \sigma_c, \alpha_c = m_c / \sigma_c$ ). Then*

$$u_c^{\text{walk}} - u_c^{\text{GC}} = f_c(0) (\mathbb{E}|\nabla z_{2,c}|^2 - \sigma_c^2) = f_c(0) \sum_a W_2(c, a) \mu_a (m_c - \mathbb{E}[z_{2,c} \mid z_{1,a} = 0]),$$

whose contact part ( $b = a$  in  $z_{2,c} = \sum_b W_2(c, b) h_{1,b}$ ) is  $f_c(0) \sum_a W_2(c, a)^2 \mu_a^2 = O(1)$ , while the exchange part ( $b \neq a$ ) is second order in  $\rho_{ab}$ .

*Proof.* Stein's identity gives  $\mathbb{E}z^2 = \mathbb{E}|\nabla z|^2 + \mathbb{E}[z \Delta z]$  for the one-homogeneous  $z = z_{2,c}$ , and  $\Delta z = \sum_a W_2(c, a) \delta(z_{1,a}) |w_a|^2$  with  $f_{1,a}(0) |w_a|^2 = \mu_a$ ; hence  $\mathbb{E}|\nabla z|^2 - \sigma^2 = m^2 - \sum_a W_2(c, a) \mu_a \mathbb{E}[z \mid z_{1,a} = 0]$ , and  $m = \sum_a W_2(c, a) \mu_a$ . Conditioning on  $z_{1,a} = 0$  sets  $h_{1,a} = 0$  (the contact) and shifts the other neurons through their correlations with  $z_{1,a}$ , which at zero mean changes their means at order  $\rho^2$ .  $\square$

Numerically (width 256,  $2^{20}$  inputs): the Gaussian closure has rms error  $1.7 \cdot 10^{-3}$ , independent walkers  $1.9 \cdot 10^{-1}$ ;  $\mathbb{E}|\nabla z|^2 - \sigma^2$  and the contact sum agree in mean (0.639 and 0.639, correlation 0.999); removing the contact term brings the walkers to within  $7.8 \cdot 10^{-4}$  of the Gaussian closure. So the Gaussian closure is the two-token walk with exact contacts. Re-projecting each layer onto a fresh one-token (Gaussian) vector with the same covariance is precisely what handles self-pinning, and the third-cumulant chain adds first-order exchange.

## 8.7 What sparsifies, and what does not

1. **Two-token objects close.** Every sum  $\sum_{l'} T_{l \leftarrow l'} X_{l'} T_{l \leftarrow l'}^\top$  accumulates into one  $n \times n$  recursion ( $n^3$  per layer): the covariance, the gradient Gram, the Gaussian closure, and any hub sum whose hub coefficient is known when the hub is born.
2. **Layer two is cheap.** The open walk of length two (the star of Theorem 8.4) is

$$\sum_a D_c(a) (W_2 G)(c, a)^2;$$

its hub coefficient is  $W_2$  itself, known at birth, so all  $n$  of them cost one product  $W_2 G$  and a Hadamard square.

3. **From layer three on the hub coefficient is a walk.** A hub born at layer  $l'$  and read at layer  $l$  needs three tokens that meet at the hub, each transported from  $l'$  to  $l$  (two arms and the reader). Nothing accumulates, and this is the third-cumulant memory. The trace over input slots (the operation the Gaussian input performs) contracts input vertices, never hidden hubs, so no operation on the input removes it.

| token-walk level       | estimator                                       | network 0, MSE                          |
|------------------------|-------------------------------------------------|-----------------------------------------|
| independent walkers    | factorised kink current                         | $O(1)$ (two layers)                     |
| + exact contacts       | Gaussian closure                                | $4.1 \cdot 10^{-6}$                     |
| + first-order exchange | third-cumulant chain (radial $\kappa_4$ off/on) | $5.6 \cdot 10^{-7} / 3.5 \cdot 10^{-8}$ |
| + cooling, $d = 2$     | Euler–Stein merge, $a \approx \frac{1}{4}$      | $6.6 \cdot 10^{-9}$                     |

## 9 Kikuchi sparsification, two dualities, and a two-tier dynamics

### 9.1 What the quantum-cut sparsification paper does

Basu, Brakensiek, Kothari and Putterman [17] sparsify the quantum-cut Hamiltonian

$$L_G = \sum_e w_e (I - A_e) \quad \text{on } \mathbb{C}^{2^{[n]}},$$

where  $A_e$  swaps the two qubits of the edge  $e$ , i.e. acts on subsets by the transposition  $(e)$ .  $L_G$  is the Laplacian of the Kikuchi (token) graph  $K_G$  at every level  $k$  at once:  $S \sim T$  iff  $|S| = |T|$  and  $S \oplus T \in E(G)$ , the symmetric exclusion process of  $k$  hard-core tokens on  $G$ . Their theorem:  $\tilde{O}(n/\varepsilon^2)$  reweighted edges preserve  $L_G$  within  $1 \pm \varepsilon$  on every state, so one sampled graph is a spectral sparsifier of every level simultaneously. Classical reductions are hopeless (the singlet projector has rank one, and CSPs with one satisfying assignment need  $n^2$  terms); the entangled structure is what makes it possible. The argument has four manoeuvres.

1. **Up and down.**  $U$  adds a token anywhere,  $D$  removes one,  $D = U^*$  and  $[D, U] = (n - 2k)I$  on level  $k$ . Harmonic functions are  $H_k = \ker D$  on level  $k$ , and  $E_k = \bigoplus_{\ell=k}^{n-k} U^{(\ell)} H_k$ . Each  $A_e$  commutes with  $U$ , so every  $L_G$  preserves every  $E_k$ , and  $L_{K_n} = k(n + 1 - k)$  on  $E_k$ .
2. **Leverage decays with the level.** On  $E_k$  an edge has  $L_e \preceq 2$  while  $L_G \succeq \lambda \frac{2m}{n^2} k(n + 1 - k)$ , so the matrix-Chernoff parameter grows like  $k \log n / \varepsilon^2$  and beats  $\log \dim E_k \lesssim k \log n$ . The importance of an edge trickles down as the level rises, and a union bound over levels closes.
3. **Up to the group, down to every level.** With  $\nabla_G = \sum_e w_e (\text{id} - (e)) \in \mathbb{C}[S_n]$  and the lift  $g(\sigma) = f(\{\sigma(1), \dots, \sigma(k)\})$ ,  $\langle f, L_e f \rangle = \langle g, \nabla_e g \rangle$ : an operator inequality proved once in the group algebra holds at all levels. Alon–Kozma give  $\nabla_G \succeq \lambda \nabla_{K_n}$  for expanders, through the octopus inequality of Caputo–Liggett–Richthammer (a star at  $u$  dominates the complete graph on its leaves weighted by  $w_{uv} w_{uw'} / \sum w$ ).
4. **From expanders to everything.** Expander decomposition, Chen’s lift of effective resistance to all levels,  $R_G(u, v) L_G \succeq L_{uv}$ , and an edge-moving lemma across weight scales. The paper’s Remark 8.5 isolates the obstruction: a bridge of a tree keeps its full importance at every level, so the decay is an average over edges, not a per-edge fact.

**Proposition 9.1** (The essence is Schur–Weyl duality).  $\mathbb{C}^{2^{[n]}} = (\mathbb{C}^2)^{\otimes n} = \bigoplus_{k \leq n/2} V_{n/2-k} \otimes S^{(n-k, k)}$ , where  $U, D$  and  $[D, U]$  generate  $\mathfrak{su}(2)$  (levels are its weight spaces),  $S_n$  permutes the

tensor factors,  $V_j$  is the spin- $j$  representation and  $S^{(n-k,k)}$  the two-row Specht module. Then  $E_k = V_{n/2-k} \otimes S^{(n-k,k)}$  and  $L_G|_{E_k} = I \otimes \rho_{(n-k,k)}(\nabla_G)$ . In particular every level inside  $E_k$  carries the same operator, which is why one sampling scheme serves all levels, and  $L_{K_n}|_{E_k} = \binom{n}{2} - c(n-k, k) = k(n+1-k)$  with  $c$  the content sum of the diagram.

*Proof.*  $H_k = \ker D_k$  is the top constituent  $S^{(n-k,k)}$  of the permutation module on  $k$ -subsets;  $U$  commutes with  $S_n$ , so  $U^{(\ell)}H_k \cong S^{(n-k,k)}$  for  $k \leq \ell \leq n-k$ , and these  $n-2k+1$  copies form one  $\mathfrak{su}(2)$  string.  $L_e = I - \rho(e)$  acts only through  $S_n$ . The sum of all transpositions acts on  $S^\lambda$  by the content sum; for  $\lambda = (n-k, k)$  it is  $\binom{n-k}{2} + \binom{k}{2} - k$ , and  $\binom{n}{2}$  minus it equals  $k(n+1-k)$ .  $\square$

## 9.2 The network has the same two-duality structure, once bosonic and once hard-core

**Proposition 9.2** (Howe duality of the Gaussian tokens and the radial foliation). *On  $L^2(\mathbb{R}^n, \gamma)$  let  $E^- = \frac{1}{2} \sum_i a_i a_i$ ,  $E^+ = \frac{1}{2} \sum_i a_i^\dagger a_i^\dagger$ ,  $H = N + \frac{n}{2}$ . Then  $[E^-, E^+] = H$ ,  $[H, E^\pm] = \pm 2E^\pm$ , and the triple commutes with  $\Gamma(O)$ ,  $O \in O(n)$ : the pair  $(O(n), \mathfrak{sl}_2(\mathbb{R}))$  is a Howe dual pair and  $L^2(\gamma) = \bigoplus_j \mathcal{H}_j \otimes M_j$ , with  $\mathcal{H}_j$  the degree- $j$  spherical harmonics (traceless token kernels,  $\ker E^-$ ) and  $M_j$  the lowest-weight  $\mathfrak{sl}_2$ -module of radial functions. The homogeneity operator is  $x \cdot \nabla = N + 2E^-$ , so the Euler–Fock ladder (Theorem 8.2) is  $E^-$ -descent:  $E^-$  removes a pair of tokens from one vertex, where the paper’s  $D$  removes one hard-core token. For a one-homogeneous  $F = r \phi(\theta)$  the mean is the  $O(n)$ -invariant component,  $u = \mathbb{E}|x| \cdot \int_{S^{n-1}} \phi d\sigma$ : in the radial foliation of  $\mathbb{R}^n \setminus 0$  the leafwise structure is  $\mathfrak{sl}_2$ , the transverse structure is  $O(n)$ , and the mean is the transverse measure of the leafwise slope.*

*Proof.*  $[a_i^2, a_i^{\dagger 2}] = 4a_i^\dagger a_i + 2$  gives  $[E^-, E^+] = N + \frac{n}{2}$ ;  $N$  changes by  $\pm 2$  under  $E^\pm$ ;  $O(n)$ -invariance of  $\sum a_i^2$  and  $\sum a_i^{\dagger 2}$  is the invariance of  $\Delta$  and  $|x|^2$ . The decomposition is the classical theory of spherical harmonics.  $x \cdot \nabla = \sum (a_i + a_i^\dagger) a_i$ . The mean formula is the polar decomposition of  $\gamma$  ( $|x|$  independent of  $\theta$ ).  $\square$

**Proposition 9.3** (Gates are hard-core tokens; a layer alternates the two dualities). *The gate pattern of a layer is a point  $S \in 2^{[n]}$ , a basis vector of  $(\mathbb{C}^2)^{\otimes n}$ , and  $g_a^2 = g_a$  is the exclusion constraint. In the gate-pair kernel this is exactly the contact term:  $\mathbb{E}[g_a g_a] = \Phi_a$ , where independent (bosonic) copies would give  $\Phi_a^2$ , so  $\text{diag}(\Phi - \Phi^2)$  is the hard-core correction. The hidden relabelling group  $S_n$  (permute the rows of  $W_l$  and the columns of  $W_{l+1}$ ) is an exact symmetry of the network, of the He law and of every chain, and acts on gate patterns as in the paper; one realisation breaks it, which is the odd sector of Proposition 5.8. Each layer maps bosonic Gaussian tokens to hard-core gates (the selection) and back to bosonic linear combinations (the next weights).*

## 9.3 What transfers: the memory has no redundancy

**Proposition 9.4** (Every hub is a bridge). *Let  $P = \sum_c P^{(c)}$  be the readout of a source as a sum of hub atoms, and let  $P_q$  keep each atom independently with probability  $q$  and weight  $1/q$ . Then*

$$\mathbb{E}\|P_q - P\|_F^2 = \frac{1-q}{q} \sum_c \|P^{(c)}\|_F^2 = \frac{1-q}{q} \frac{\|P\|_F^2}{\rho_{\text{eff}}}, \quad \rho_{\text{eff}} = \frac{\|\sum_c P^{(c)}\|_F^2}{\sum_c \|P^{(c)}\|_F^2}.$$

*On network 0,  $\rho_{\text{eff}} = 0.998$ – $1.004$  at ages 1, 3, 6 (stage-9 workflow, lens 4), so the relative error of hub sampling is  $(1-q)/q$  whatever  $n$ . In the paper’s terms every hub atom is a bridge (their Remark 8.5): its importance does not decay, because the readout sees one level only and the atoms are orthogonal there.*

*Proof.* The atoms are kept independently, so  $\mathbb{E}\|P_q - P\|^2 = \sum_c \text{Var}(\xi_c/q) \|P^{(c)}\|^2$  with  $\xi_c \sim \text{Bernoulli}(q)$ .  $\square$

The level decomposition that does exist on the Gaussian side (Proposition 9.2) concentrates the mean in the single sector  $j = 0$ , the opposite of the spreading that sparsification needs. The manoeuvre that does transfer is the third: prove in the group, descend to every level. For the network the group is the hidden relabelling  $S_n$ , under which the ensemble-averaged chain is symmetric; whatever is proved there holds for all  $k$ -hub joint cumulants at once, and the realisation-specific (odd) part is what lies outside.

## 9.4 A higher tier with random dynamics

**Proposition 9.5** (Localization is a stochastic flow on the first-layer weights). *An input  $x \sim N(m, \Sigma)$  is the standard input through the effective first layer  $W_1 \Sigma^{1/2}$  with bias  $W_1 m$ . Along an adapted localization with control  $C_t$ ,  $d\Sigma_t = -\Sigma_t C_t \Sigma_t dt$  and  $dm_t = \Sigma_t C_t^{1/2} dB_t$ : the effective weights contract deterministically and the bias is a matrix-weighted Brownian martingale. Klartag's process is the dual flow (the metric is random, the centre fixed).*

**Theorem 9.6** (Gate martingales, wall currents and covariance consumption). *For every layer  $l$  and neuron  $a$ ,  $p_t(l, a) = \mathbb{P}_{\nu_t}(z_{l,a} > 0)$  is a martingale with*

$$dp_t(l, a) = \langle J_t(l, a), \Sigma_t C_t^{1/2} dB_t \rangle, \quad J_t(l, a) = \mathbb{E}_{\nu_t}[\delta(z_{l,a}) \nabla z_{l,a}],$$

the wall current: the Gaussian surface integral of the wall's normal under the current ball. The conditional gate covariance  $R_t = \mathbb{E}_{\nu_t}[gg^\top] - p_t p_t^\top$ , over all layers jointly, satisfies

$$\mathbb{E} R_t = R_0 - \mathbb{E} \int_0^t \Gamma_s ds, \quad \Gamma_s = J_s^\top \Sigma_s C_s \Sigma_s J_s \succeq 0, \quad R_\infty = 0,$$

so  $\text{Cov}(\text{gates}) = \mathbb{E} \int_0^\infty \Gamma_s ds$ : the contact entries  $\Phi - \Phi^2$  and the exchange entries of the Kikuchi weights, within and across layers, are integrated Grams of wall currents. The currents trickle up the layers,  $J(l, a) = \sum_b W_l(a, b) \mathbb{E}_\nu[\delta(z_{l,a}) g_{l-1,b} \nabla z_{l-1,b}]$ .

*Proof.*  $\pi(m, \Sigma) = \mathbb{E} g(m + \Sigma^{1/2} Z)$  solves  $\partial_\Sigma \pi = \frac{1}{2} \nabla_m^2 \pi$ , so the Itô drift  $\langle \partial_\Sigma \pi, d\Sigma \rangle + \frac{1}{2} \langle \nabla_m^2 \pi, d\langle m \rangle \rangle$  vanishes, and  $\nabla_m \pi = \mathbb{E}[\delta(z) \nabla z]$  in the sense of distributions.  $\mathbb{E}[gg^\top \mid \mathcal{F}_t]$  is a martingale and  $d(pp^\top) = dp p^\top + p dp^\top + d\langle p, p^\top \rangle$  with  $d\langle p, p^\top \rangle = \Gamma dt$ . Completeness of the localization and  $g^2 = g$  give  $R_\infty = 0$ . The recursion is the chain rule for  $\nabla z_{l,a}$ .  $\square$

At layer one along Eldan's path the identity is Price's (Plackett's) formula integrated along the correlation time  $s = t/(1+t)$ :  $\text{Cov}(g_a, g_b) = \int_0^1 \rho_{ab} \varphi_2(\alpha_a, \alpha_b; \rho_{ab}s) ds$ , checked to six digits for centred and non-centred pairs and for  $\rho = 1$  (the contact term). This is the dynamical counterpart of Klartag's contacts. Walls whose current is non-negligible under the current ball are in contact. The ball's gate covariance is consumed by them at the rate of their current Gram, and a wall that leaves contact for good freezes its gate. An adaptive control  $C_t$  (Lovett–Meka, Klartag) steers which walls are consumed when. By the defect-path theorem the total error of any estimator is path-independent, but its distribution along the path is not.

**Theorem 9.7** (Localization resolves deep gates first). *Two copies  $X, X'$  that are conditionally independent given  $\mathcal{F}_t$  have input correlation  $t/(1+t)$ . In the infinite-width limit the expected fraction of layer- $l$  gates on which they disagree is  $\theta_{l-1}(t)/\pi$ , where  $\theta_0 = \arccos \frac{t}{1+t}$  and  $\cos \theta_{k+1} = J(\theta_k)/\pi$ .*

Equivalently the neuron-averaged unresolved gate variance is  $\mathbb{E}\text{Var}(g_{l,a} \mid \mathcal{F}_t) = \theta_{l-1}(t)/(2\pi)$ . It decreases with depth, so localization freezes the deep gates before the shallow ones, strongly early ( $\theta_0$  large) and weakly late ( $\theta_l \approx \theta_0 - l\theta_0^2/3\pi$ ).

*Proof.*  $\mathbb{P}(g(X) \neq g(X')) = 2\mathbb{E}[p_t(1-p_t)]$  for conditionally independent Bernoulli( $p_t$ ) copies. Over the neuron ensemble the pair  $(z_{l,a}(X), z_{l,a}(X'))$  is centred Gaussian with correlation  $\cos \theta_{l-1}$  (the dual-kernel recursion of Theorem 2.4), and two centred Gaussians of correlation  $\cos \theta$  have different signs with probability  $\theta/\pi$ .  $\square$

| measured / predicted       | layer 1   | 2         | 4         | 8         | 12        | 16        |
|----------------------------|-----------|-----------|-----------|-----------|-----------|-----------|
| $t = 1$ ( $\rho = 0.5$ )   | .333/.333 | .287/.292 | .223/.236 | .174/.173 | .130/.138 | .136/.115 |
| $t = 9$ ( $\rho = 0.9$ )   | .140/.144 | .135/.136 | .116/.124 | .106/.105 | .082/.092 | .098/.081 |
| $t = 99$ ( $\rho = 0.99$ ) | .044/.045 | .042/.044 | .041/.043 | .040/.041 | .033/.039 | .043/.037 |

(Fraction of disagreeing gates; He network of width 512, 64 pairs; the last layer drifts at finite width, as in Proposition 8.5.)

## 9.5 The two tiers form a non-commuting square

On  $L^2$  of the input and the observation noise there are two filtrations. The depth tower  $\mathfrak{B}_1 \subset \dots \subset \mathfrak{B}_L$  consists of the tile algebras, generated by the gates of layers  $\leq l$  (the lower tier, the Kikuchi vertex sets). The localization filtration  $\mathfrak{F}_t = \sigma(y_s, s \leq t)$  is the upper tier. Each family is commutative and nested.

**Proposition 9.8** (Non-commuting square and the angle operator). *For  $0 < t < \infty$  and  $l \geq 1$  the pair  $(\mathfrak{B}_l, \mathfrak{F}_t)$  is not a commuting square:  $E_{\mathfrak{F}_t}$  maps a gate indicator to a smooth function strictly between 0 and 1, which is not  $\mathfrak{B}_l$ -measurable, so  $E_{\mathfrak{B}_l}E_{\mathfrak{F}_t} \neq E_{\mathfrak{F}_t}E_{\mathfrak{B}_l}$  and the von Neumann algebra generated by the two filtrations is noncommutative. The angle operator  $\Theta_{l,t} = E_{\mathfrak{B}_l}E_{\mathfrak{F}_t}E_{\mathfrak{B}_l}$  is the noise-stability Markov operator of the depth- $l$  tiling at correlation  $t/(1+t)$ ; its two-point marginals are the time-correlated Kikuchi weights  $K_{l,t}(a,b) = \mathbb{P}(g_{l,a}(X) = 1, g_{l,b}(X') = 1)$ , which run from  $\Phi_a\Phi_b$  at  $t = 0$  (independent walkers) to  $K_l$  at  $t = \infty$  (contacts and exchange on). At layer one  $K_{1,t}$  is the tetrachoric series with  $\rho_{ab}$  replaced by  $\rho_{ab}t/(1+t)$ , the diagonal included ( $\rho_{aa} = 1$ ), so the contact term is the resummed series at  $t = \infty$ . The gate process along a localization is non-Markov exactly because the square does not commute.*

*Proof.*  $\mathbb{E}[\mathbf{1}\{z > 0\} \mid y_t] = \Phi(\cdot)$  of the conditional mean and variance, smooth and in  $(0,1)$ . For  $f, g \in L^2(\mathfrak{B}_l)$ ,  $\langle f, \Theta g \rangle = \mathbb{E}[\mathbb{E}(f \mid \mathfrak{F}_t)\mathbb{E}(g \mid \mathfrak{F}_t)] = \mathbb{E}f(X)g(X')$  with  $X, X'$  conditionally independent given  $\mathfrak{F}_t$ , whose correlation is  $\text{Var } \mathbb{E}(X \mid y_t) = t/(1+t)$ . The tetrachoric expansion of  $\Phi_2$  gives the layer-one formula.  $\square$

Taken together: the lower-tier Kikuchi weights, conditioned on a Gaussian input law  $N(m, \Sigma)$ , form a field over the upper tier that solves the heat equation, and localization paths are its Brownian characteristics. A moment chain defines a field that is not heat-flat. Its residual is the Euler–Stein defect, and the defect-path theorem says that the error is the residual integrated along the characteristics. This fibred picture ties both tiers together. Its leaf spaces are Hausdorff (the sphere for the radial foliation, the space of balls for localization), so the noncommutativity of the framework lies in the non-commuting square of Proposition 9.8, not in a pathological leaf space. Whether the depth tower at infinite depth, with tail equivalence of gate sequences, yields a genuinely noncommutative tail algebra is open.

## 10 What the placement explains, what it predicts, and what is open

### 10.1 What the placement explains

1. **Why the mean is hard and the densities are easy.** The mean is a sum over walls of a density at the kink times a conditional transport on the wall (Theorem 4.1). Densities are concentration-function coefficients and the Gaussian surrogate gets them right; the conditional transports are rare-event-relative objects (conditioning on a measure-zero wall), and every chain’s error lives there.
2. **Why the third-cumulant memory has an identity leg.** It is the label of the kink at which the cumulant was born (Theorem 5.3); the memory is the record of all pinnings, transported.
3. **Why it cannot be compressed.** Kinks have small codegree at He initialisation, so their atoms are orthogonal; the frame bound (Theorem 5.4) makes the memory exactly as compressible as its transport leg, and the complementarity of the neuron algebra with the modular frame of the layer state (Theorem 5.7) makes state-respecting (spectral) compressions blind to the hub diagonal up to  $2\sqrt{2/n}$ . The same small codegree that makes Sahasrabudhe’s nibble concentrate makes our compression fail.
4. **Why ensemble surrogates fail.** The memory’s readout is odd in the last weight matrix (Proposition 5.8); free and NNGP descriptions are the symmetric phase.
5. **Where the Euler–Stein constants come from.** The correction averages the defect over Gaussian balls of uniform radius (Theorem 6.1); a defect of Hermite contact order  $d$  decays like  $r^{d+1}$ , so  $a = 1/(d+2)$ , and  $d$  is the order of the concentration coefficient the estimator gets wrong (Theorem 6.2, Proposition 6.3); the three measured constants fit  $d = 0, 0, 2$ .
6. **Why depth behaves algebraically.** Lengths are preserved and chords contract like  $3\pi/l$  (Theorem 2.4) because the activation’s heat trace is critical at linear order and has a  $\beta^{3/2}$  kink term (Proposition 6.6).

### 10.2 What it predicts (falsifiable, theoretical)

1. The defect of the exact first-order chain along Eldan’s path decays like  $r^3 = (1+\tau)^{-3/2}$  (order  $d = 2$ ), not like  $(1+\tau)^{-1}$ ; a chain that also corrects the curvature coefficient  $f''(0)$  at the kinks has a defect of order  $d \geq 4$  and merge constant at most  $1/6$ .
2. On the official networks, the number of walls crossed by a great-circle arc grows linearly in depth (about  $n\theta_0/\pi$  per layer) while the number of gates that differ between its endpoints grows like  $3n \log L$ .
3. The best hub-index compression of any old source has the error of the low-rank truncation of its transport leg  $T_{l \leftarrow l'} C$  (Theorem 5.4 with a near-isotropic Hadamard leg); a tier cannot do better than that number, which can be computed without running any chain.
4. The complementarity constant of the layer state at depth is  $\approx 2\sqrt{2/n}$  up to the collective mode, so every state-preserving coarse-graining of a layer onto spectral data retains at most a fraction  $\approx 8/n$  of the centred energy of any neuron-diagonal observable (a gate pattern, a hub diagonal).

### 10.3 Open problems

1. **Contact order at depth.** Prove Conjecture 6.4: show that the defect of a chain whose first neglected term is of Hermite order  $d$  at every wall has the profile  $r^{d+1}$  along the localization, despite the nonlinear dependence of the downstream pre-activation means on the localization centre.
2. **A contact-resolved estimator.** The kink-current formula asks for the on-wall transports  $\mathbb{E}[\partial_{h_i} h_L e_a \mid z_{l,a} = 0]$  for all walls. To first order these are pinning responses (Lemma 5.2); their sum over walls is one directional derivative of the downstream chain per output, which forward mode computes with an identity leg (the memory) and reverse mode with one adjoint per output. Either way the cost is that of the memory. Is there an order-two quantity (a pinning response of pinning responses) whose sum over walls collapses, as the radial fourth-cumulant scalar does for the fourth cumulant?
3. **The induced Dirac ensemble.** Compute the spectral dimension and spectral variance [7] of the selected Dirac operators  $P_x D P_x$  and of the transports  $T_{l \leftarrow l'}$  by free multiplicative convolution, and compare with the measured stable ranks 164, 92, 65, 47, 29, 23 of the cocycle at ages 1, 2, 3, 4, 6, 8; decide whether the emergence of the collective eigenvalue of the layer covariance (from 0.9% to 37–48% of the energy) is a Baik–Ben Arous–Péché transition in depth, the analogue of the one-cut to two-cut transition of [5].
4. **Bootstrap with kink generators.** Is the island of the moment problem at low degree narrow at large  $n$  because of a factorisation analogous to large- $N$  factorisation [6]? If so, positivity would certify chains cheaply.
5. **Klartag in reverse.** Klartag’s ellipsoid stops when about  $d^2/2$  contacts hold its  $d(d+1)/2$  shape parameters. The marginal value of a source age decays by a factor 0.75 per age and about eleven ages matter, roughly  $4n$  effective contacts held in memory. Is there a counting argument that fixes this number from the dimension of the third-cumulant readouts that the mean actually uses?

### Verification

Four scripts in `scripts/` compare the two sides of each identity numerically:

- `verify_stage10.py` (output in `notes/stage10/verify_stage10.txt`);
- `verify_tokens.py` and `verify_tokens_contacts.py` (Section 8);
- `verify_kikuchi.py` (Section 9).

They run no estimator and no network of the competition.

| statement                                              | check                                                                                         | result                                                                                                            |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Theorem 4.1 (kink current)                             | $\mathbb{E}h_L$ against the wall sum with a $\pm 0.05$ window, $3 \cdot 10^6$ inputs, $L = 3$ | $n = 16$ : agreement to noise (1.5%); $n = 6$ : 25% off, dead tiles carry 16%                                     |
| Theorem 2.2 (Crofton)                                  | crossings of $2 \cdot 10^4$ random walls by a random curve in $\mathbb{R}^{50}$               | 31.28 against $\Lambda/\pi = 31.39$                                                                               |
| Theorem 2.4,<br>Proposition 6.6                        | iterate the angle map                                                                         | $\theta_l l / 3\pi = 0.90, 0.985, 0.992$ at $l = 10^2, 10^3, 2 \cdot 10^3$ (log corrections)                      |
| Proposition 2.3 (unit gain)                            | tangent and norm gains at $\sigma_w^2 = 2$ , $n = 400$                                        | 1.009, 1.002                                                                                                      |
| Proposition 6.5 (Mehler, Hermite coefficients of relu) | closed form against the Hermite series                                                        | equal to $10^{-8}$                                                                                                |
| Theorem 6.2 (contact order)                            | stage-9 $\tau$ -integral for $d = 0, 2, 4$                                                    | 0.497, 0.246, 0.164 against $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}$ (truncated tail)                              |
| Theorem 5.3 (births are contacts)                      | $\kappa_3/\varepsilon^2$ by Sobol quasi-Monte Carlo, $2^{23}$ points                          | ratio to the hub formula 1.064, 1.030, 1.015, 1.009 at $\varepsilon = 0.16, \dots, 0.02$                          |
| Theorem 5.7 (complementarity)                          | $\ E_B E_A - J\ $ against $\ C\ $ for $S = WW^\top$                                           | equal; 0.333 / 0.173 against $2\sqrt{2/n} = 0.354$ / 0.177 ( $n = 64, 256$ )                                      |
| Theorem 5.4 (frame bound)                              | 2000 random rank-10 compressions                                                              | bound holds                                                                                                       |
| Proposition 8.7 (contact term)                         | two layers, width 256, $2^{20}$ inputs                                                        | walkers $1.9 \cdot 10^{-1}$ , closure $1.7 \cdot 10^{-3}$ ; contact sum explains the gap (corr 0.999, slope 0.99) |
| Theorem 8.4 (walk sums, contacts)                      | $\kappa_3$ split into self, contact and distinct parts, widths 24, 96                         | distinct part 0.059, 0.0115 against leading diagrams 0.064, 0.0133                                                |
| Proposition 8.5 (token number)                         | $\mathbb{E} \nabla z ^2 / \text{Var } z$ on a width-256 He network                            | matches $1/(1 - \cos \theta)$ to layer 13; 10.6 against 13.0 at layer 16                                          |
| Proposition 9.1 (Schur–Weyl)                           | $\binom{n}{2}$ minus the content sum against $k(n+1-k)$                                       | equal for $n = 8, 12$ , all $k$                                                                                   |
| Theorem 9.6 (covariance consumption)                   | layer one, Eldan’s path, four mean/correlation pairs including $\rho = 1$                     | equal to six digits                                                                                               |
| Theorem 9.7 (deep gates first)                         | disagreement of conditionally independent copies, width 512                                   | table in Section 9; within noise to layer 12                                                                      |

The asymptotic  $3\pi/l$  is reached slowly ( $O(\log l/l)$  corrections), so at the depth  $L = 16$  of the competition the exact recursion, not its asymptotic form, is the statement to use.

## References

- [1] Stage 9 note: *the mean of a random ReLU network as a central localization* (this repository, [notes/stage9](#)).
- [2] Stage 8 note: *Localization in the commutant: modular tilts, Kikuchi–Toeplitz quantization and transverse measures*.
- [3] J. Sahasrabudhe, Probabilistic combinatorics at exponentially small scales, arXiv:2512.15077.

- [4] B. Klartag, Lattice packing of spheres in high dimensions using a stochastically evolving ellipsoid, arXiv:2504.05042.
- [5] M. D’Arcangelo, S. Gnutzmann, Symmetry breaking and phase transitions in random non-commutative geometries and related random-matrix ensembles, arXiv:2601.14141.
- [6] M. Khalkhali, N. Pagliaroli, Bootstrapping noncommutative geometry with Dirac ensembles, arXiv:2512.08694.
- [7] J. W. Barrett, P. Druce, L. Glaser, Spectral estimators for finite non-commutative geometries, arXiv:1902.03590.
- [8] J. W. Barrett, L. Glaser, Monte Carlo simulations of random non-commutative geometries, J. Phys. A 49 (2016) 245001.
- [9] R. Wang, L. C. Lau, H. Zhou, Derandomizing matrix concentration inequalities from free probability, arXiv:2601.08111.
- [10] A. Sah, J. Sahasrabudhe, M. Sawhney, On the Spielman–Teng conjecture, arXiv:2405.20308.
- [11] Y. Cho, L. K. Saul, Kernel methods for deep learning, NeurIPS 2009.
- [12] B. Poole, S. Lahiri, M. Raghu, J. Sohl-Dickstein, S. Ganguli, Exponential expressivity in deep neural networks through transient chaos, NeurIPS 2016.
- [13] S. Hayou, A. Doucet, J. Rousseau, On the impact of the activation function on deep neural networks training, ICML 2019.
- [14] B. Hanin, D. Rolnick, Complexity of linear regions in deep networks, ICML 2019.
- [15] H. Scarf, A min-max solution of an inventory problem, in *Studies in the Mathematical Theory of Inventory and Production*, Stanford 1958.
- [16] D. Nualart, G. Peccati, Central limit theorems for sequences of multiple stochastic integrals, Ann. Probab. 33 (2005) 177–193; I. Nourdin, G. Peccati, *Normal Approximations with Malliavin Calculus*, Cambridge 2012.
- [17] A. Basu, J. Brakensiek, P. K. Kothari, A. Putterman, Quantum cut sparsifiers, arXiv:2606.09728.
- [18] J.-B. Lasserre, Global optimization with polynomials and the problem of moments, SIAM J. Optim. 11 (2001) 796–817.
- [19] M. Takesaki, Conditional expectations in von Neumann algebras, J. Funct. Anal. 9 (1972) 306–321.
- [20] M. A. Rieffel, Metrics on state spaces, Doc. Math. 4 (1999) 559–600.
- [21] A. Connes, *Noncommutative Geometry*, Academic Press 1994.